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LECTURE · MATE6301 NANOMATERIALS

Photodetectors

From photons to signals, and how we see the nanostructure behind them

Light–matter interaction · junctions · figures of merit · SEM, EDX, TEM, XRD · porous semiconductors · A case study in plasma defect engineering

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Roadmap: from photon to signal to structure

ROADMAP · 90 MINUTES · 5 BLOCKS

A 0–35 MIN

Photodetector essentials

B 35–60 MIN

Seeing the nanostructure: SEM, EDX, TEM, XRD

C 60–68 MIN

Porous semiconductors

D 68–84 MIN

Case study: plasma domain engineering

E 84–90 MIN

Summary and discussion

THROUGHOUT

18 live simulations · 4 checkpoints · figures from the case-study paper

make the material → **see** its structure (SEM · EDX · TEM · XRD) → link structure to **performance**

A Photodetector essentials

0–35 MINUTES

Photon energy and absorption, why semiconductors, the p–n junction, and the figures of merit that decide whether a detector is any good.

Why light–matter interaction matters

A · THE CONVERSION CHAIN

Sensing

UV index, flame and gas detection, ambient light, pulse oximetry.

Si, GaN, Ga_2O_3, organic photodiodes

Imaging

CMOS cameras, thermal imaging, X-ray and medical imaging, astronomy.

Si pinned photodiodes, InGaAs, HgCdTe, microbolometers

Communication

Fibre links at 1310/1550 nm, LiFi, free-space optics, LiDAR.

InGaAs p–i–n and APDs, Ge-on-Si, Si SPADs

Energy & catalysis

Photovoltaics and photo(electro)catalysis share the same first steps.

absorb → separate → collect (or react)

photon → absorption → e^-/h^+ pair → separation → collection → current

A photodetector makes each step as efficient, fast and quiet as the application needs. Every step depends on the **material's structure**: band gap, crystallinity, defects and surface area. That is why the second half of this lecture is about seeing that structure.

Photon energy decides who can absorb it

A · SIMULATION 1

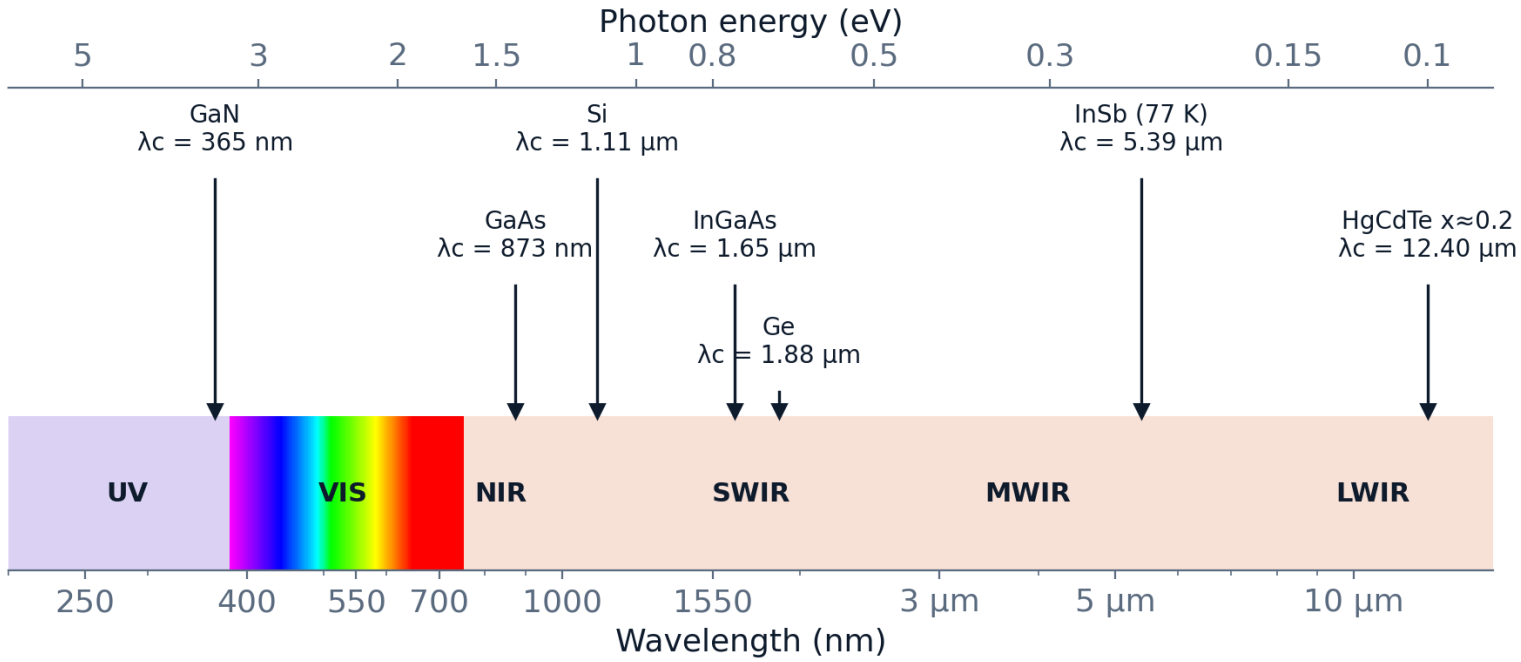
$$E = h\nu = hc/\lambda$$

$$E\ [\text{eV}] \approx \mathbf{1240} / \lambda\ [\text{nm}]$$

$$\lambda_c\ [\mu\text{m}] = 1.24 / E_g\ [\text{eV}]$$

A semiconductor absorbs a photon across its gap only if **$h\nu \geq E_g$**. That one inequality explains why silicon cameras are blind at 1550 nm.

Anchors: 400 nm = 3.10 eV · 850 nm = 1.46 eV · 1550 nm = 0.80 eV · 10 µm = 0.124 eV



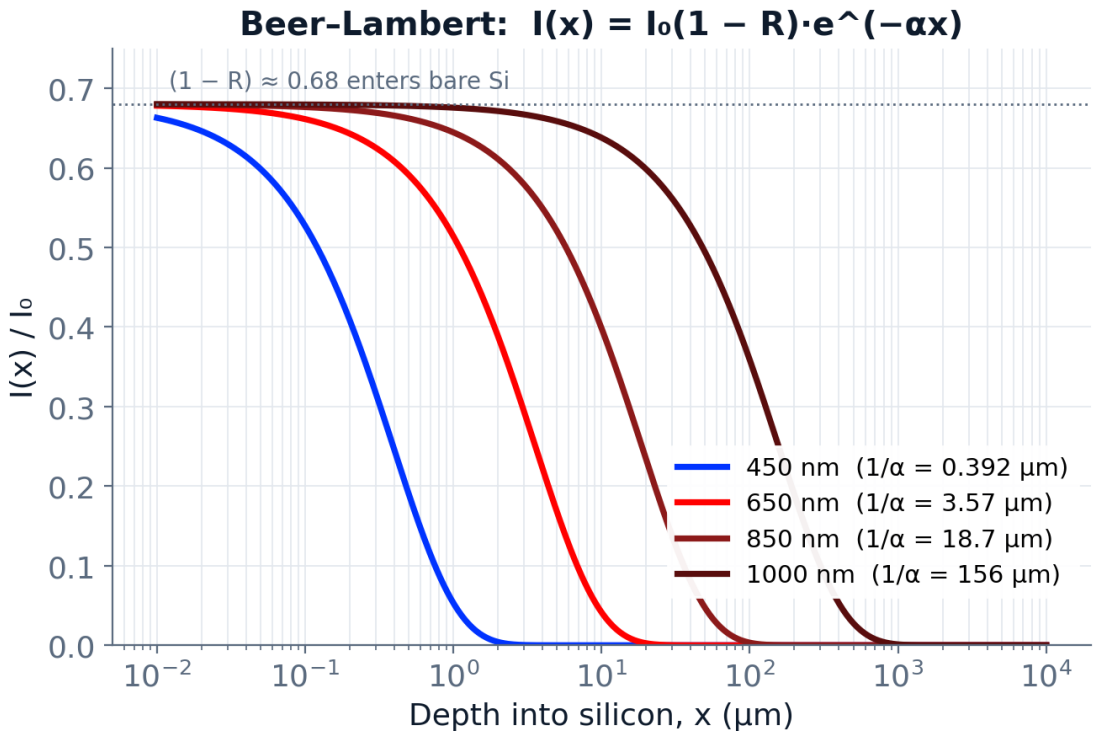
Reflection and Beer–Lambert: how deep does light go?

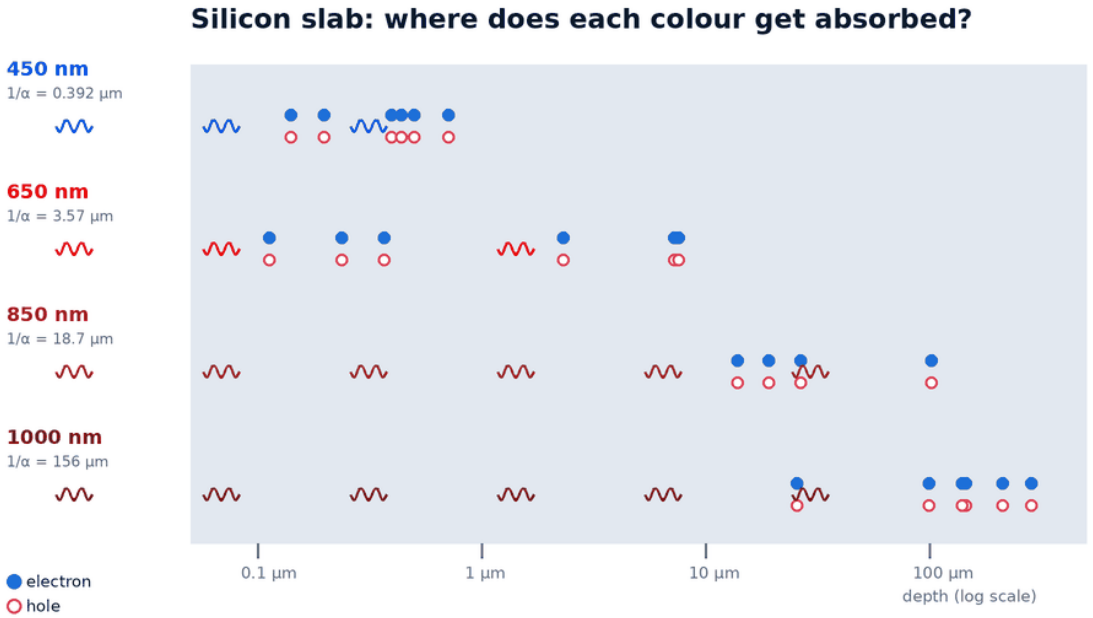
A · SIMULATION 2

$$R = [(n-1)^2 + \kappa^2] / [(n+1)^2 + \kappa^2]$$

$$I(x) = I_0(1-R)\cdot e^{-\alpha x}$$

$$\eta_{abs} = (1-R)(1-e^{-\alpha d})$$

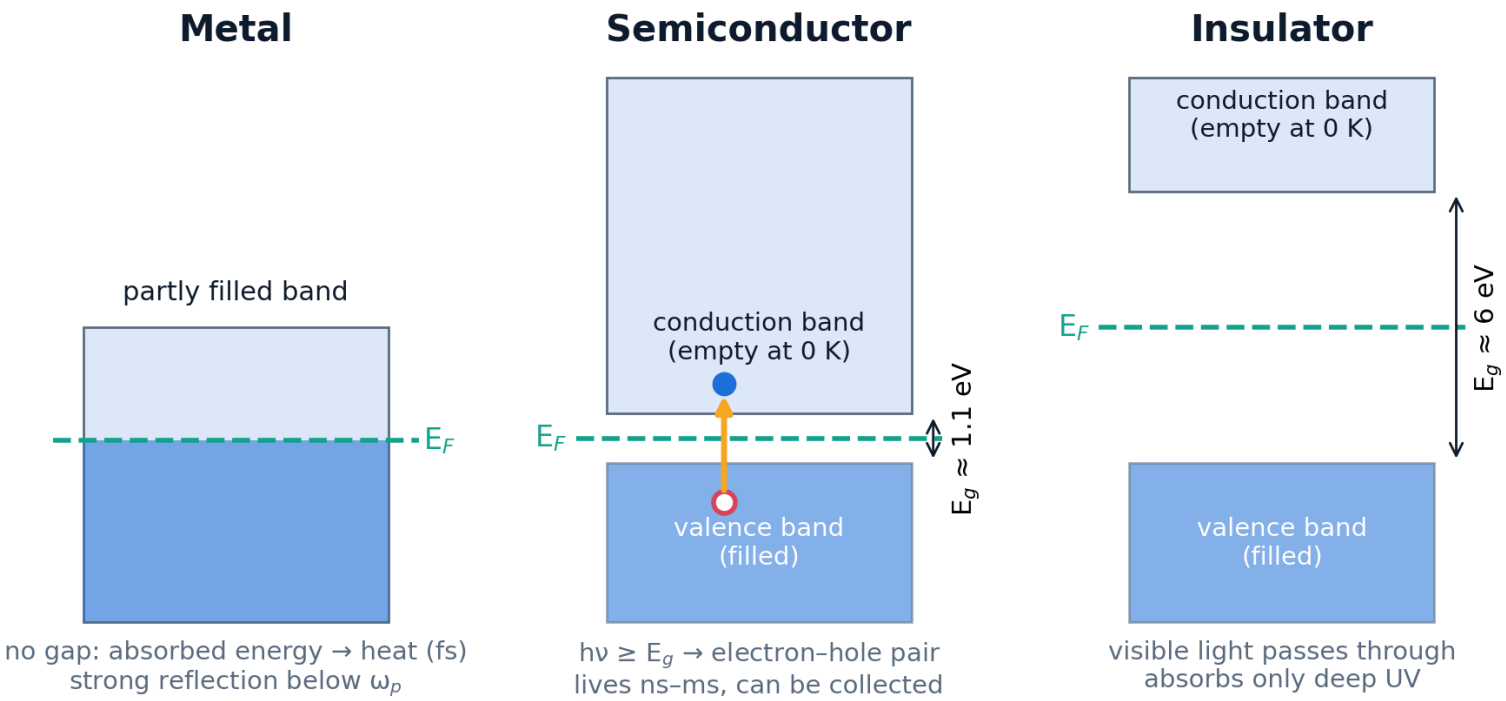




Bare Si reflects ≈ 32 % (n ≈ 3.6) before any absorption. Animation: blue light stops within 1 μm; 1000 nm light needs hundreds of μm.

Metal vs semiconductor vs insulator under light

A · SIMULATION 3



Drude: $\varepsilon(\omega) = \varepsilon_\infty - \omega_p^2 / (\omega^2 + i\gamma\omega)$

Copper chalcogenides such as the CuTe in our case study are semiconductors that absorb across the visible. Ultrawide-gap materials (diamond, AlN, Ga_2O_3) are used for solar-blind UV detection.

Metal

E_F inside a band: reflects below ω_p; absorbed energy → heat in fs.

Semiconductor

Gap 0.1–3.5 eV, controllable doping, ns–ms lifetimes: the sweet spot.

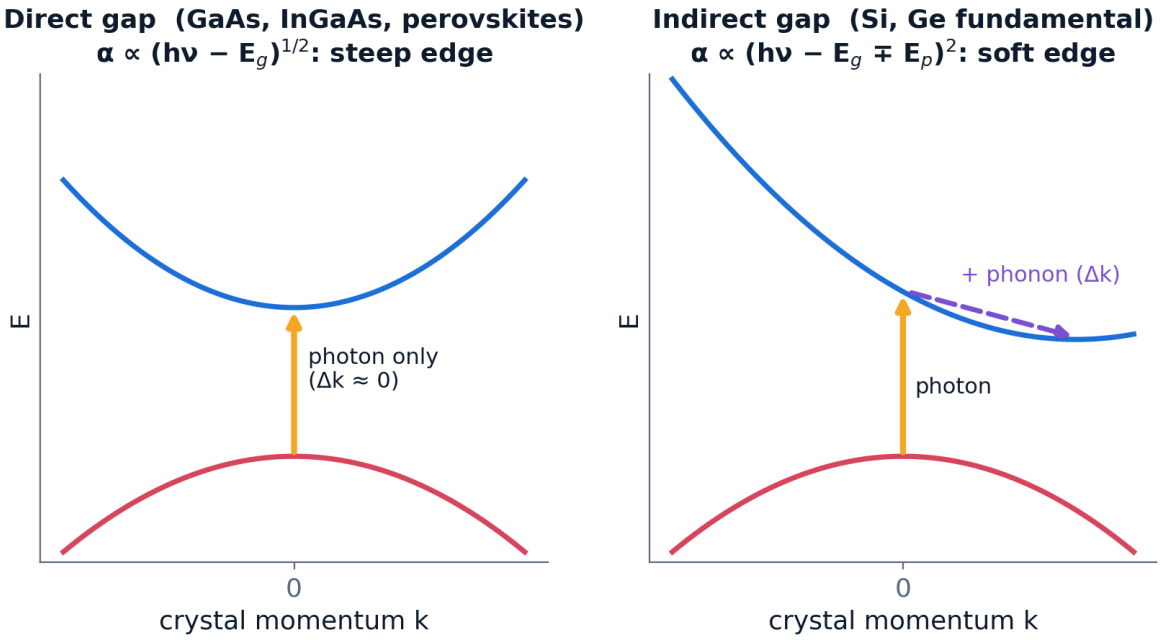
Insulator

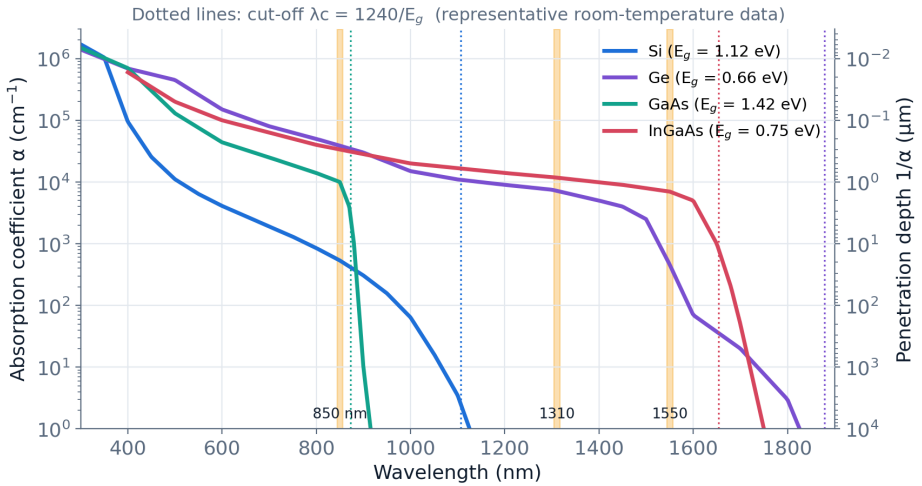
Gap > 5 eV ($SiO_2 \approx 9$ eV): transparent until deep UV.

Direct vs indirect gaps and the absorption edge

A · SIMULATION 4

direct: $\alpha \propto (h\nu - E_g)^{1/2}$ · indirect: $\alpha \propto (h\nu - E_g \mp E_p)^2$





Direct (GaAs, InGaAs, perovskites)

Photon alone does it: strong absorption just above E_g.

Indirect (Si)

A phonon supplies Δk: weak edge, thick absorbers.

Edges are blurred by **Urbach tails** from disorder. The Urbach energy becomes a key number in the case study: it tracks how disordered the pore walls are.

Choosing a material is choosing a spectral window

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A · MATERIAL TABLE

Material	E_g (eV)	Type	λ_c	Typical band	Notes
AlN	~6.0	direct	207 nm	deep UV	AlGaN alloys for solar-blind
β-Ga_2O_3	~4.9 (4.4–4.9)	direct/near-direct	253 nm	solar-blind UV-C	intrinsically solar-blind
GaN	3.4	direct	365 nm	UV-A	visible-blind
4H-SiC	3.26	indirect	380 nm	UV	robust, low dark current
ZnO	3.37	direct	368 nm	UV	persistent photoconductivity
$MAPbBr_3$	~2.3	direct	540 nm	visible	narrowband single-crystal demos
MoS_2 (1L)	~1.85	direct	670 nm	visible	bulk ~1.2 eV, indirect
$MAPbI_3$	~1.55	direct	800 nm	vis–NIR	tunable by halide and A-site
GaAs	1.42	direct	873 nm	vis–NIR	high α, fast
Si	1.12	indirect	1.11 µm	0.2–1.1 µm	CMOS workhorse
$In_{0.53}Ga_{0.47}As$	0.75	direct	1.65 µm	SWIR 0.9–1.7 µm	telecom; lattice-matched to InP
Ge	0.66 (direct 0.80)	indirect	1.88 µm	NIR/SWIR	higher dark current
PbS CQD	~0.7–1.5	direct	tunable	NIR/SWIR	size-tunable; CQD-on-CMOS
Black phosphorus	~0.3 (bulk)	direct	4.1 µm	MIR–vis	polarisation-sensitive
InSb (77 K)	0.23	direct	5.4 µm	MWIR 3–5 µm	cooled
HgCdTe x≈0.2 (77 K)	~0.1	direct	~12 µm	LWIR 8–14 µm	alloy-tunable
Graphene	0	semimetal	none	UV–THz	2.3 % absorption per layer

Rule of thumb: pick the smallest-gap material that still covers your wavelength. A smaller gap means more intrinsic carriers, more dark current and more cooling.

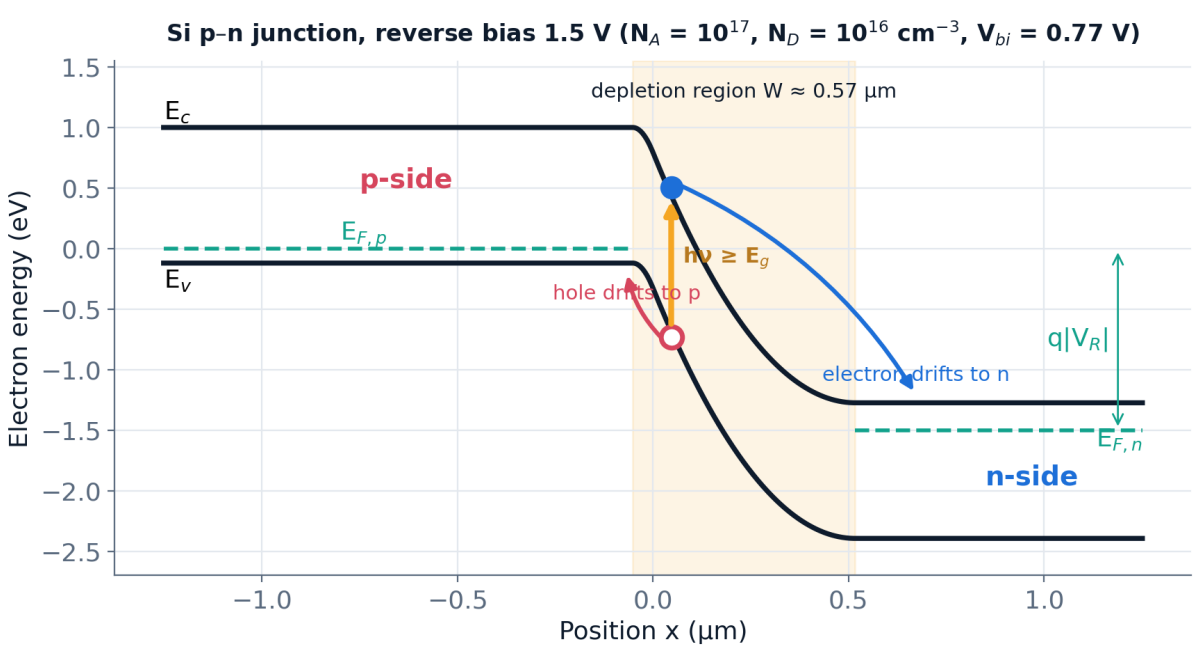
The p–n junction: a built-in field separates the pair

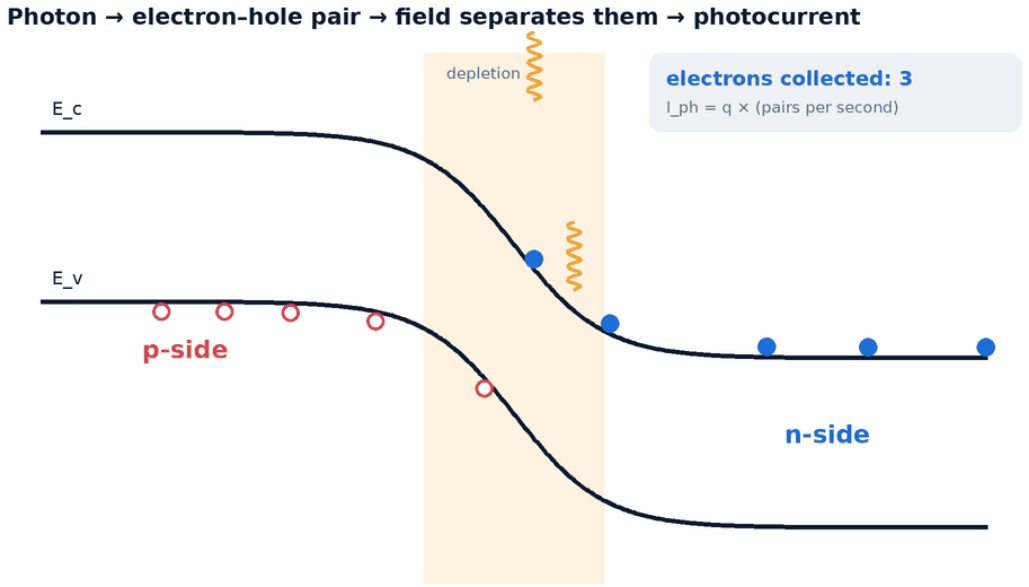
A · SIMULATION 5

$$V_{bi} = (kT/q)\cdot\ln(N_A N_D/n_i^2)$$

$$W = \sqrt{[2\varepsilon_s(V_{bi} - V)(N_A + N_D)/(qN_A N_D)]}$$

$$L = \sqrt{(D\tau)}$$



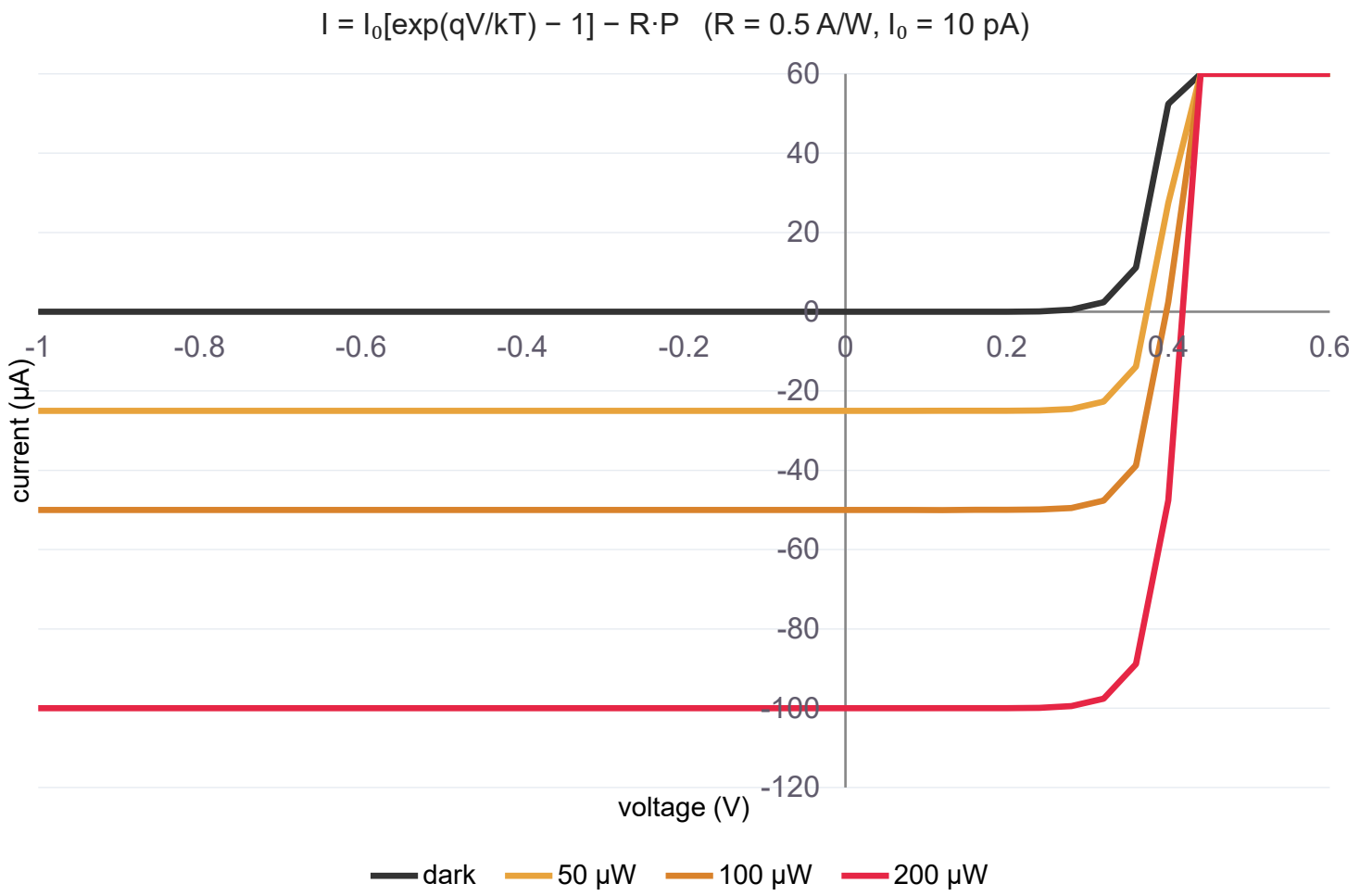


Pairs generated in the depletion region are swept apart in ps–ns; pairs generated further away must diffuse a distance L before recombining.

In a mesoporous film every wall is only 10–20 nm thick: carriers are always close to a surface.

The illuminated diode: dark and light I–V in four quadrants

A · SIMULATION 6



$$I = I_0[e^{qV/nkT} - 1] - I_{ph}$$

$I_{sc} = I_{ph}$ (linear) · $V_{oc} \propto \ln(I_{ph}/I_0)$ (log)

III · Photoconductive mode

Reverse bias: linear, fast, more dark current.

IV · Photovoltaic mode

Power out: solar cells, self-powered detectors.

The case-study films were tested by recording current with light chopped on and off.

Photoconductors: gain from trapped carriers

A · SIMULATION 7

$$\Delta\sigma = q(\Delta n \cdot \mu_n + \Delta p \cdot \mu_p)$$

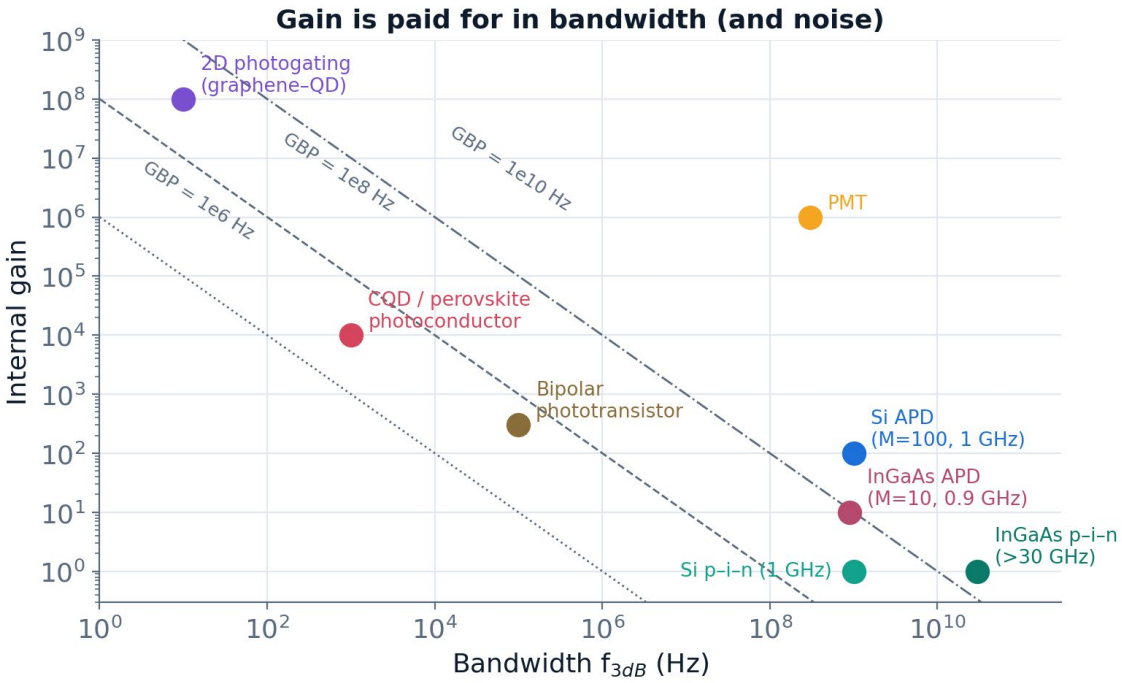
$$G = \tau / \tau_{tr}, \quad \tau_{tr} = L^2 / (\mu V)$$

If one carrier is trapped for τ while the other crosses in $\tau_{tr} \ll \tau$, the contacts re-inject carriers: each photon moves many electrons.

Bandwidth ≈ 1/(2πτ), so G × bandwidth ≈ $1/(2\pi\tau_{tr})$ is fixed. **Traps give gain and cost speed.**

Surface traps in nanostructures (pores, grain boundaries, dangling bonds) are exactly these traps. Passivating them speeds up the response, as the case study shows.

Worked: L = 10 µm, µ = 100 cm²/Vs, V = 1 V, τ = 1 ms → τ_{tr} = 10 ns, G = 10^5, f_{3dB} ≈ 160 Hz



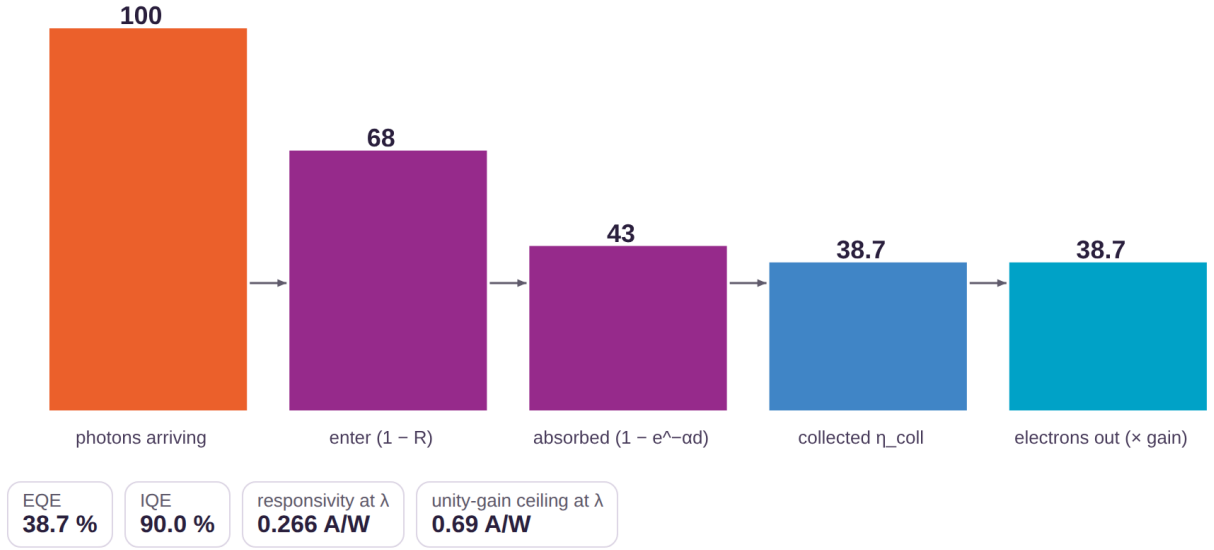
Figures of merit 1: quantum efficiency and responsivity

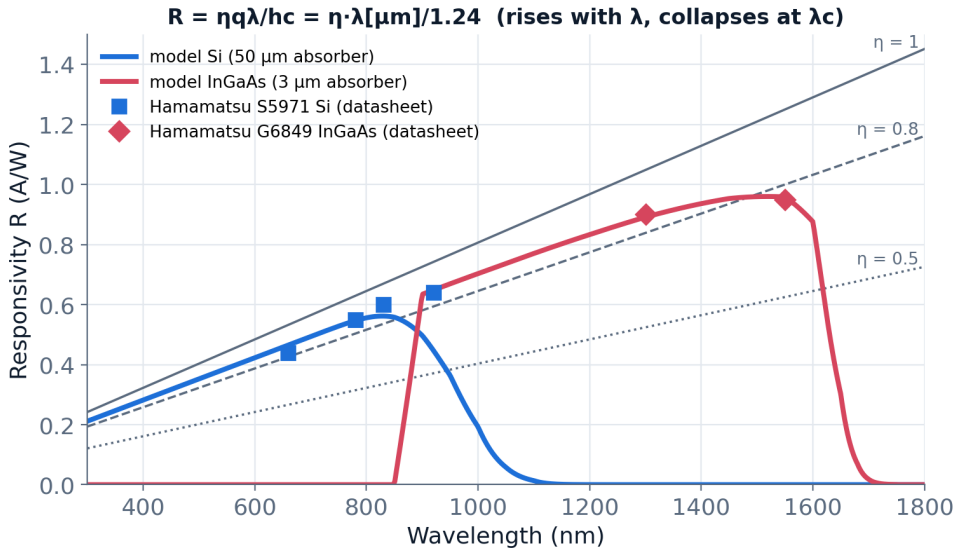
A · SIMULATION 8

$EQE = (I_{ph}/q) / (P/hv)$

$EQE = (1 - R)(1 - e^{-\alpha d}) \cdot \eta_{coll} \cdot (G)$

$R = I_{ph}/P = \eta \cdot \lambda[\mu m]/1.24 \ \ (A/W)$





Worked: Si at 850 nm, η = 0.8 → R = 0.8 × 0.85 / 1.24 = **0.55 A/W**. Commercial unity-gain photodiodes sit at η ≈ 0.76–0.86: a sanity anchor when a paper claims thousands of A/W.

Figures of merit 2: noise, NEP and detectivity

A · SIMULATION 9

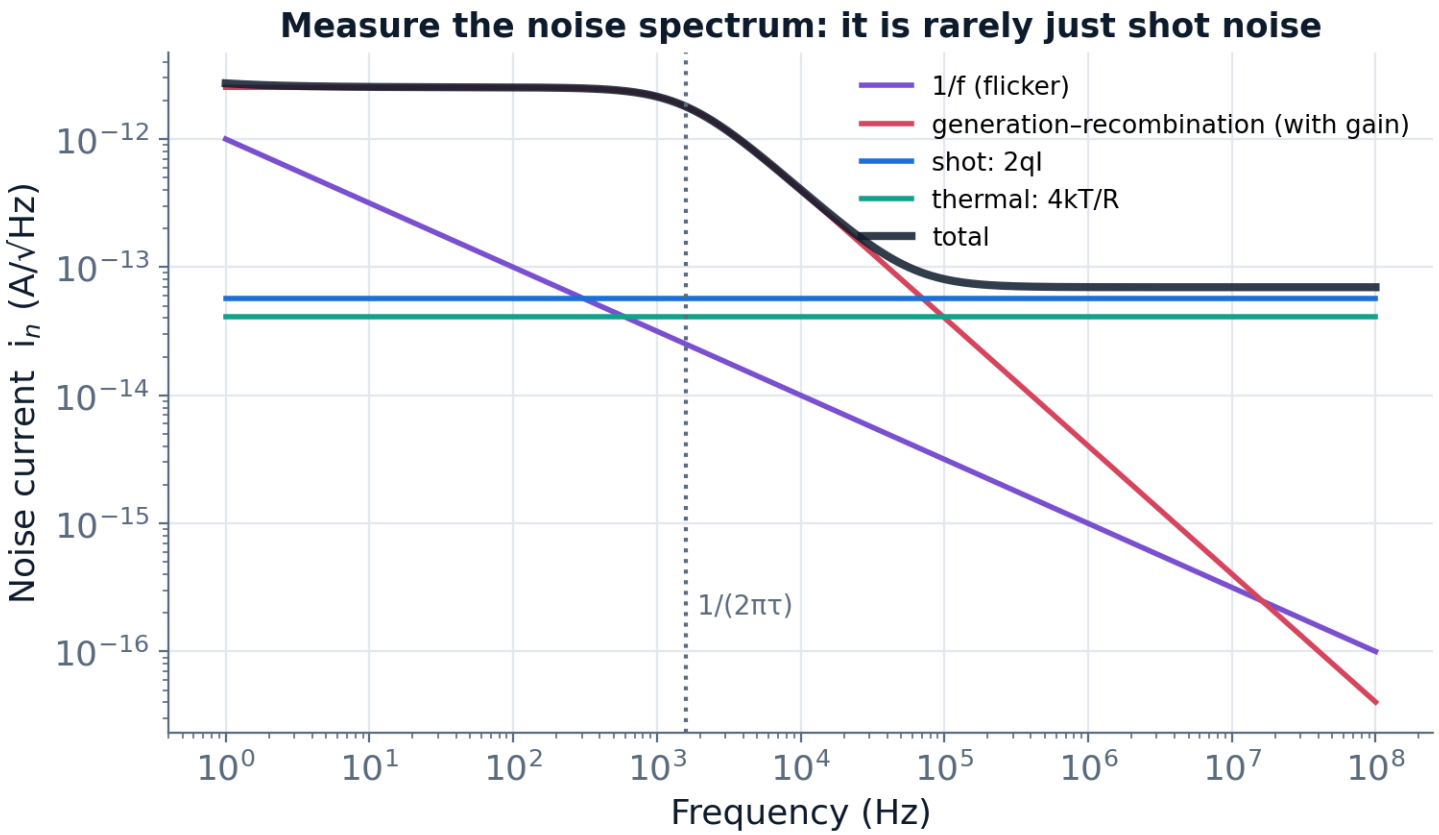
Shot: $\langle i^2 \rangle = 2qI\Delta f$ · Thermal: $4kT\Delta f/R$

$NEP = i_n / R$ (W/√Hz)

$D^* = \sqrt{(A\Delta f)} / NEP$ (Jones)

Trap states add generation–recombination and 1/f noise. A "D*" calculated only from dark current ignores them and can overestimate detectivity by orders of magnitude. **Measure the noise.**

Hamamatsu S5971 Si PIN: I_d = 0.07 nA, R = 0.64 A/W → NEP ≈ 7.4 × 10^{-15} W/√Hz, D^* ≈ 1.4 × 10^{13} Jones.



Device types and trade-offs

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A · FILTER BY NEED

Device	Internal gain	Typical bandwidth	Noise	Bias	Typical applications
Photoconductor	1–10^8 (traps)	Hz–MHz	g–r + 1/f, high dark current	required	cheap IR (PbS), UV (ZnO), nanostructured films
p–n photodiode	1	kHz–100 MHz	shot + thermal; lowest at 0 V	0 V or reverse	light meters, PPG
p–i–n photodiode	1	100 MHz to >50 GHz	shot + thermal	reverse, few V	fibre receivers, LiFi
Schottky / MSM	1	GHz to 100s GHz	higher dark current	reverse / bias	UV (GaN, Ga_2O_3), ultrafast
APD	10–1000	100 MHz to 25+ GHz	shot × M^2F	tens–hundreds V	LiDAR, telecom
Phototransistor	10^2–10^8	Hz–MHz	gain-amplified; 1/f	required	ambient sensing, 2D research
Photoelectrochemical (PEC)	1	Hz–kHz	electrolyte, surface states	0 V or small	self-powered sensing, photocatalysis
Thermal (bolometer)	none	ms	thermal fluctuation	bias / none	thermal imaging, THz

Every gain mechanism has a bandwidth bill and a noise bill. Nanostructured and porous films usually sit in the photoconductor / PEC rows: their surfaces and grain boundaries decide gain, speed and noise.

B Seeing the nanostructure

35–60 MINUTES

Once a sample is made: SEM for morphology, EDX for composition, TEM for nanoscale and atomic structure, XRD for phase and crystallinity.

Every technique answers a different question

SEM

What does it look like?

Morphology, pore size, film thickness, particle shape.

EDX

What is it made of?

Elements present, atomic ratios, elemental maps.

TEM

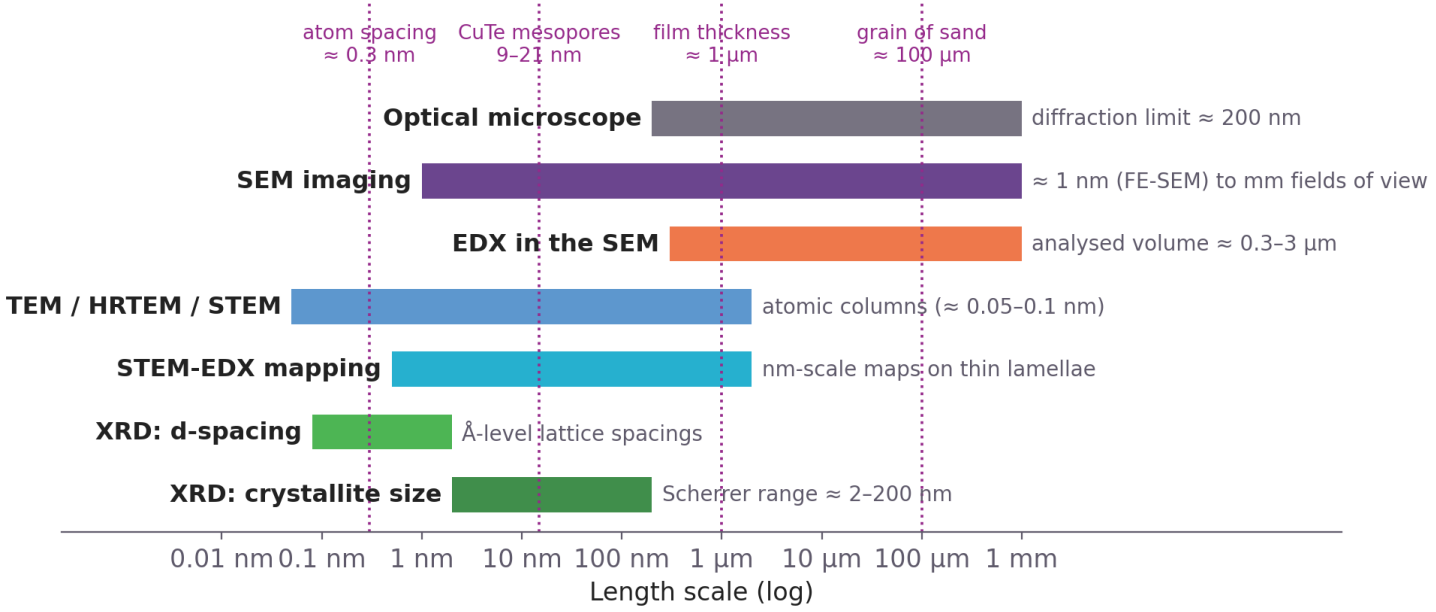
How is it arranged inside?

Grains, lattice fringes, d-spacings, defects, cross-sections.

XRD

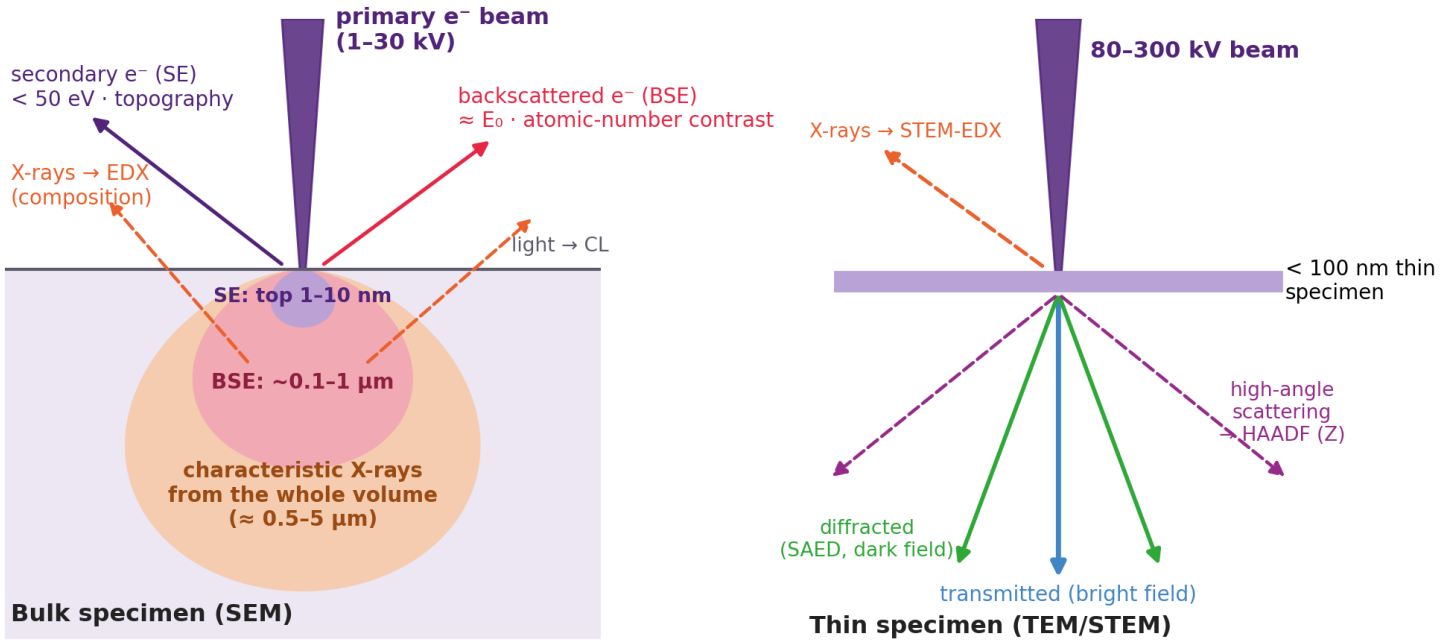
Which crystal, how crystalline?

Phase ID, crystallite size, strain, texture.



Electron–matter interaction: one beam, many signals

B · THE PHYSICS BEHIND SEM, EDX AND TEM



Secondary electrons (SE)

< 50 eV, escape from the top 1–10 nm → surface topography (SEM).

Backscattered electrons (BSE)

Energy ≈ E_0, from ~0.1–1 µm → atomic-number (Z) contrast.

Characteristic X-rays

From the whole interaction volume → composition (EDX).

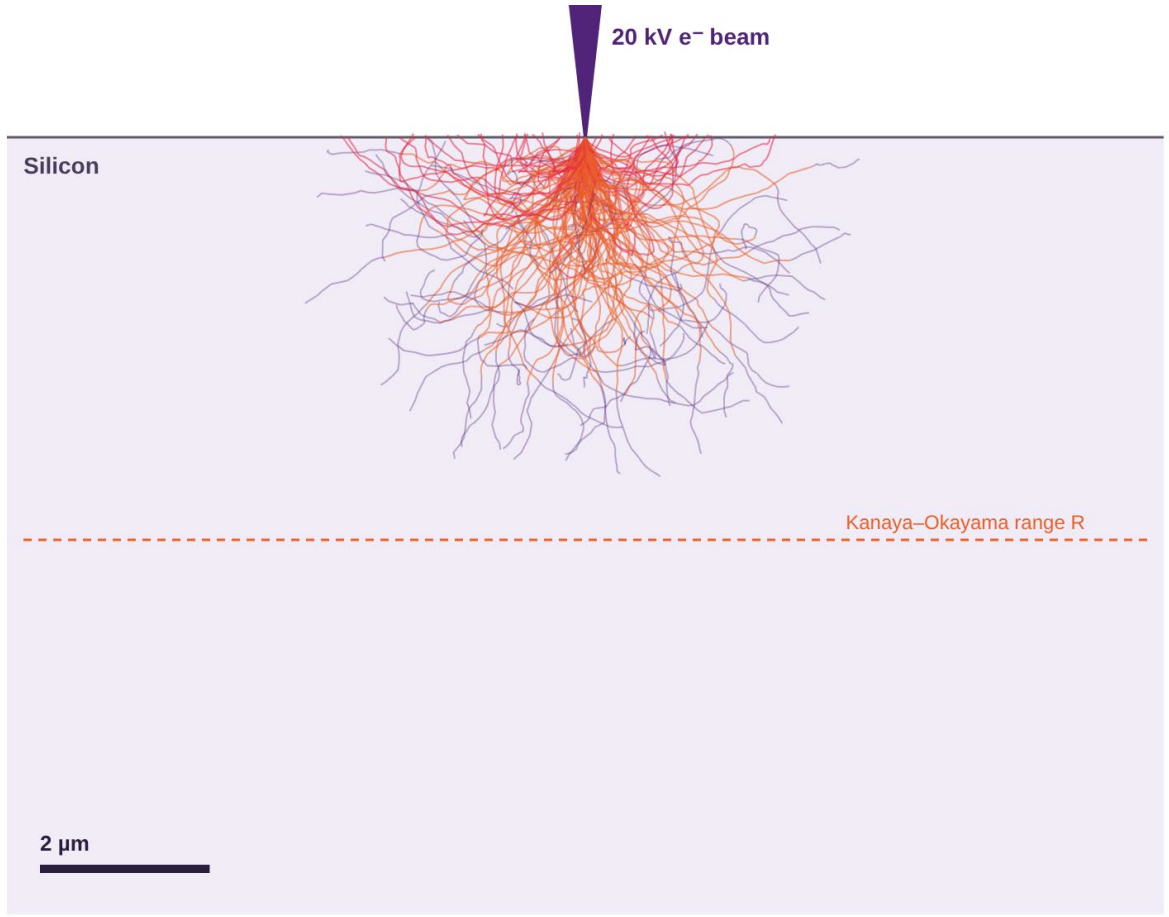
Thin specimen (TEM)

Transmitted, diffracted and high-angle scattered electrons → images, SAED, HAADF.

Kanaya–Okayama range: $R\ [\mu m] = 0.0276 \cdot A \cdot E_0^{1.67} / (Z^{0.889} \cdot \rho)$ → at 20 kV: Si ≈ 4.7 µm · Cu ≈ 1.5 µm · Au ≈ 0.86 µm

Simulation: how big is the interaction volume?

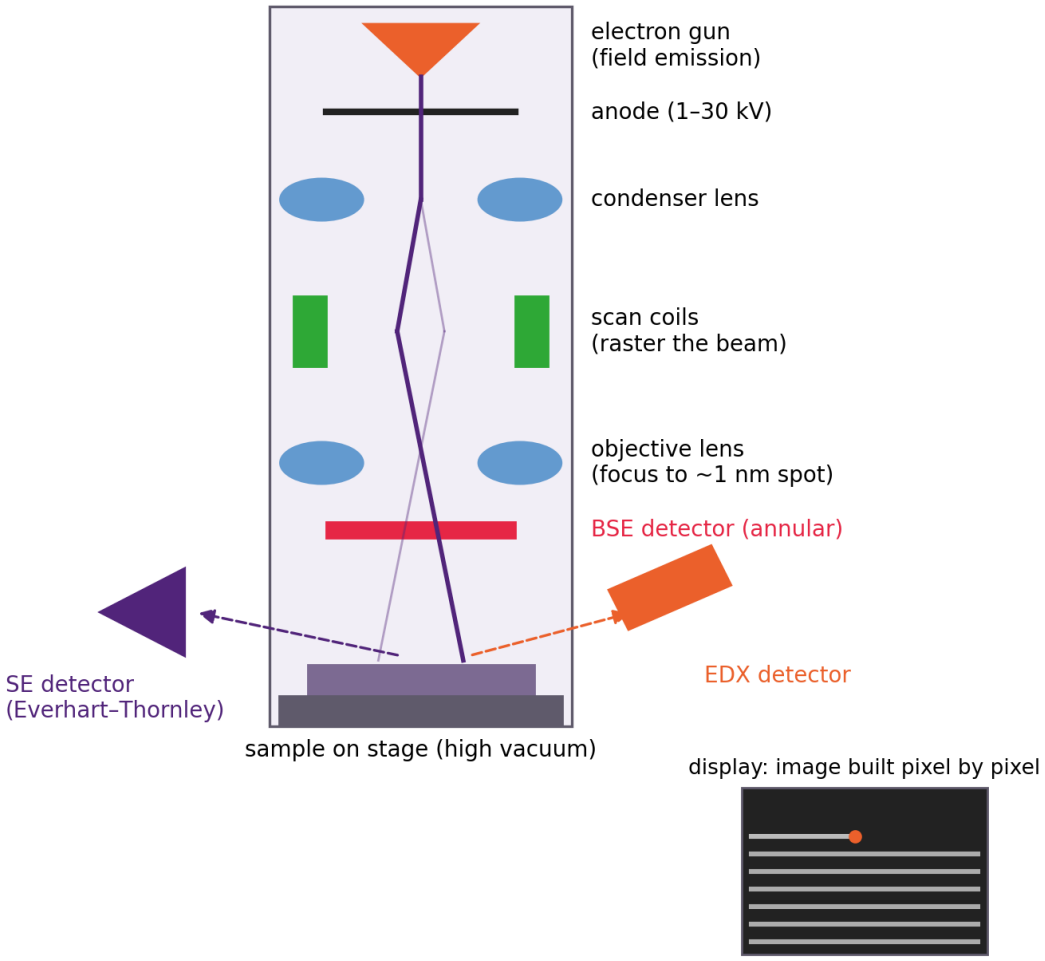
B · MONTE CARLO ELECTRON TRAJECTORIES



- **Si at 20 kV**: a pear-shaped volume ~4–5 µm deep.
- **Au**: under 1 µm, and many more electrons bounce back out (BSE): heavy elements look bright.
- **5 kV**: the volume shrinks dramatically → true surface detail.
- **Cu K edge highlight**: below ~12 kV almost no trajectory stays above 8.98 keV, so Cu Kα is barely excited.

E_0 in keV, A in g/mol, ρ in g/cm³ for the range formula.

SEM: a focused beam, scanned point by point



Electron source

Field-emission gun, accelerating voltage 1–30 kV, high vacuum.

Lenses and scan coils

Condenser and objective lenses focus a ~1 nm spot; scan coils raster it across the sample.

Detectors

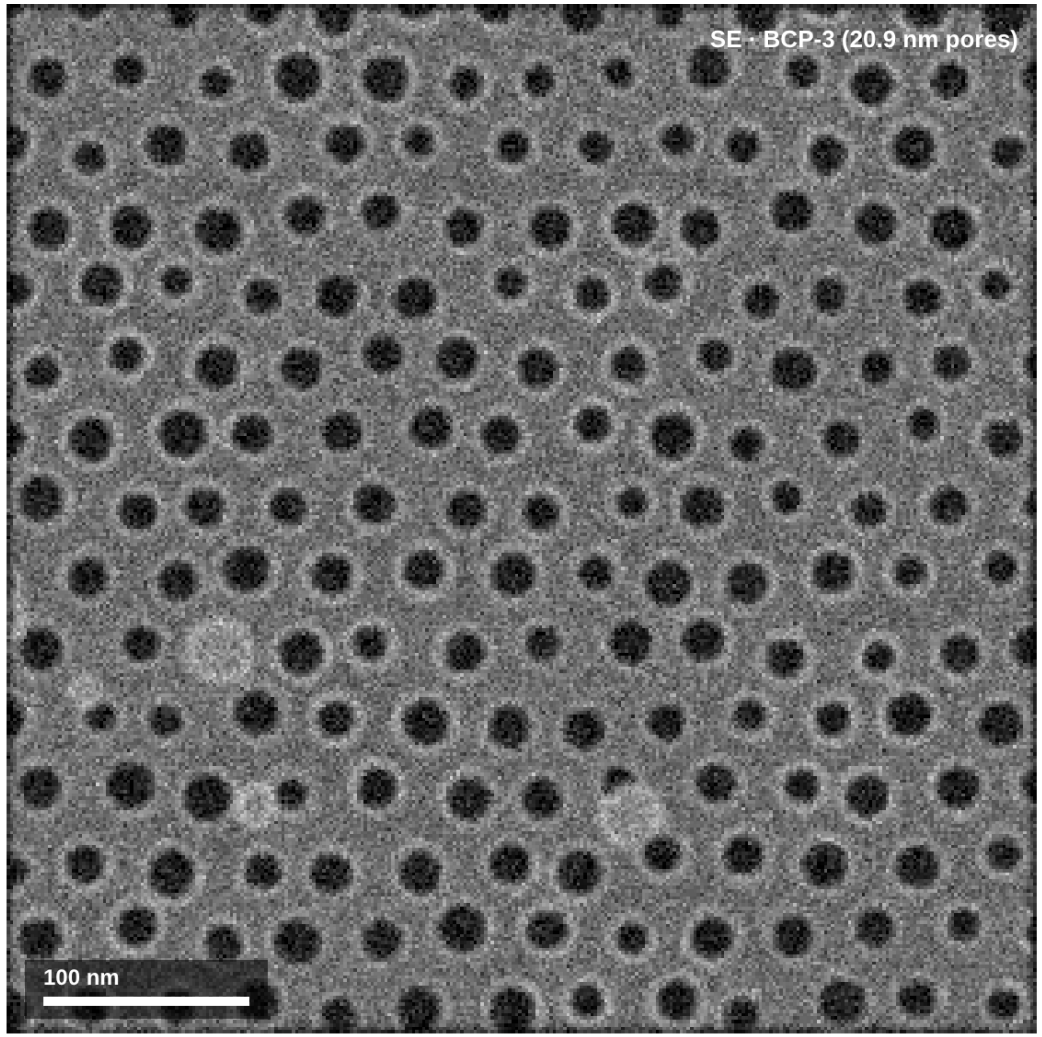
SE (Everhart–Thornley or in-lens), BSE (annular, under the pole piece), EDX (X-rays).

Image

Each pixel = detector signal at that spot. Magnification = display width / scanned width. Resolution is set by spot size and interaction volume, not magnification.

Simulation: build an SEM image of a mesoporous film

B · SIMULATION 11 · RASTER SCAN, SPOT SIZE, DWELL TIME, DETECTOR



- Start with **BCP-3** at 500 nm field of view.
- Enlarge the **spot** past the pore size: pores vanish at the same magnification.
- Shorten the **dwell time**: noise grows.
- Switch to **BSE**: Au particles light up (higher Z); edge contrast weakens.
- Zoom out to **3 µm**: 9 nm pores are lost once pixel > pore.

SEM in practice: getting a trustworthy image

SE vs BSE

SE (< 50 eV, top 1–10 nm): edges and topography. Use for pore shape and size.
BSE (≈ E_0): brighter for heavier elements. Use for phase contrast and particles.

Choosing the voltage

Low kV (1–5 kV): small interaction volume, true surface detail, less charging.
High kV (15–20 kV): better signal on conductive samples; needed to excite K-lines for EDX.

Charging and coating

Insulators accumulate charge: bright streaks, drifting images.
Fixes: thin Pt/Au or C coating, low kV, low-vacuum mode, conductive tape to ground.

Measuring pores

Threshold the image (e.g. ImageJ/Fiji), measure ≥ 100 pores, report mean ± SD and the histogram, not one hand-picked pore.

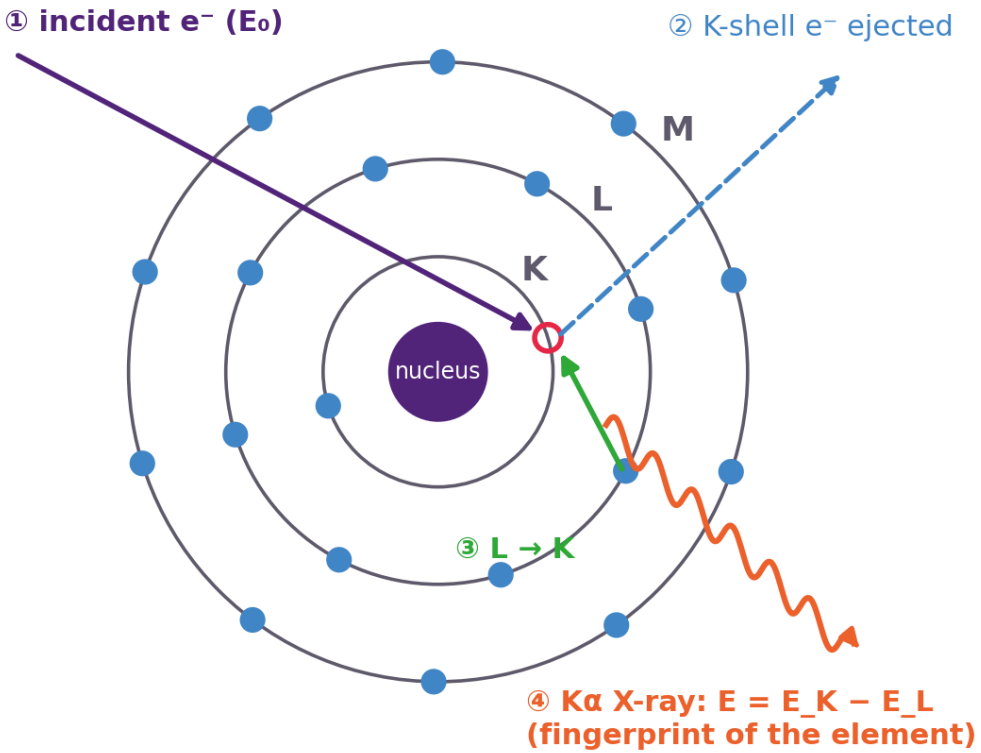
Cross-sections

Tilt or fracture the sample (or FIB-cut it) to measure film thickness and see whether pores run through the film.

Limits

Surface only. Tells you nothing about crystal structure. Beam damage on polymers and soft materials.

EDX: characteristic X-rays fingerprint the elements



① – ④ The process

The beam ejects an inner-shell (K) electron; an outer electron (L) fills the vacancy; the energy difference leaves as an X-ray.

$E(K\alpha) = E_K - E_L$ · Moseley: $\sqrt{\nu} \propto (Z - \sigma)$

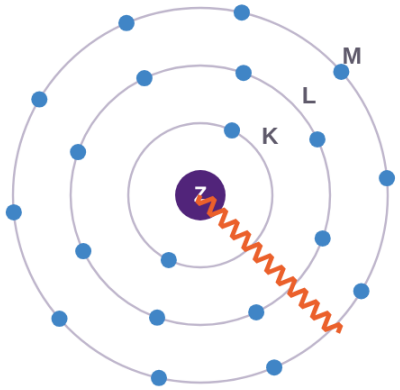
Lines you will see for CuTe on Au

Cu Kα 8.05 keV · Cu Lα 0.93 keV · Te Lα 3.77 keV · Au Mα 2.12 keV (substrate) · O Kα 0.52 keV

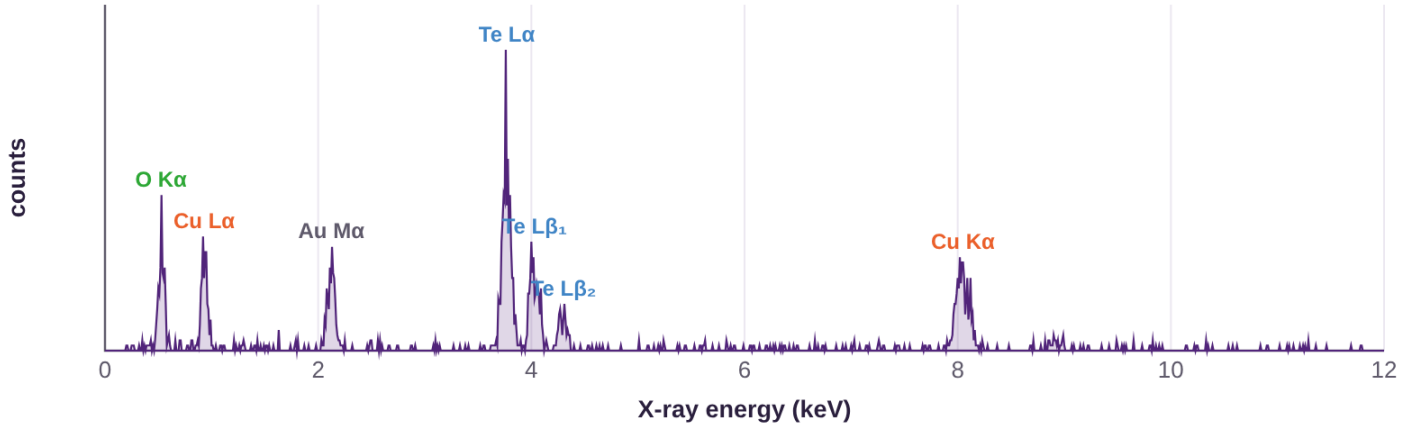
Overvoltage

$U = E_0 / E_{edge}$ should be ≥ 1.5–2. Cu K edge = 8.98 keV, so Cu Kα needs ≳ 15 kV.

Simulation: watch an EDX spectrum build up

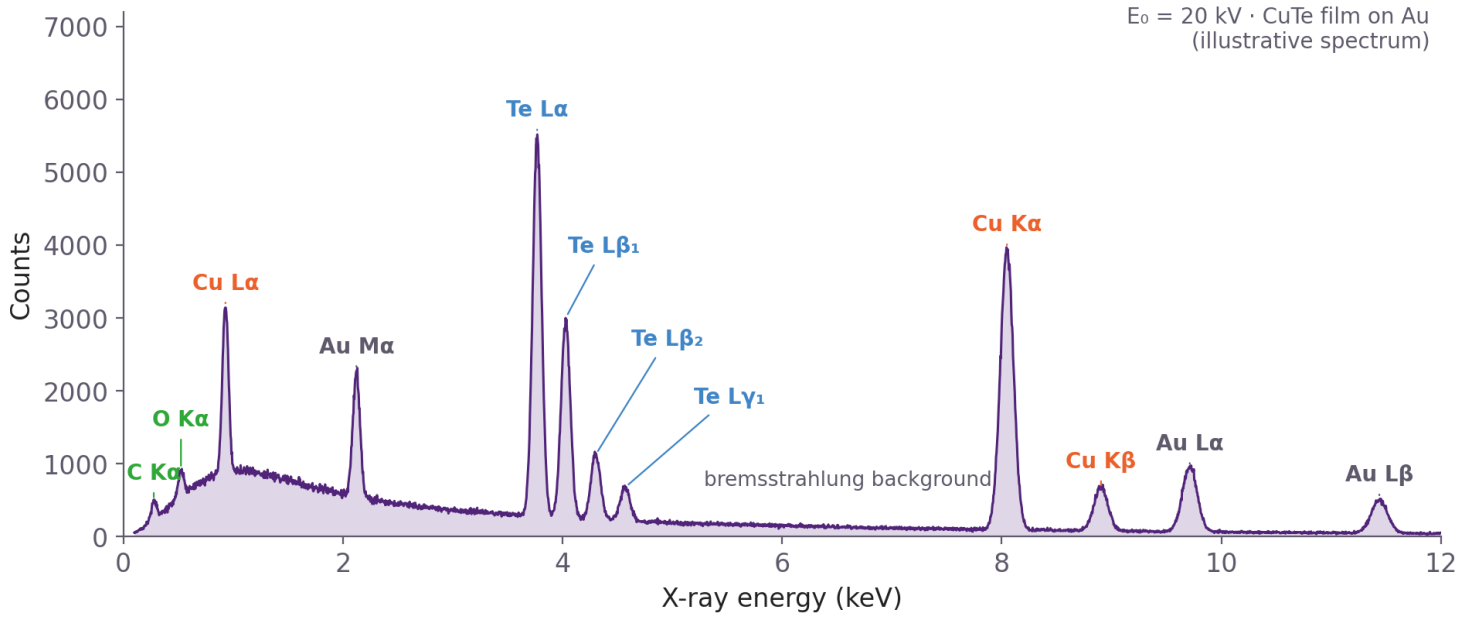


Kα X-ray → detector



- Set Cu : Te = 50 : 50 at 20 kV and let counts accumulate: the measured ratio converges; error falls as 1/√counts.
- Drop to **10 kV**: Cu Kα disappears (10/8.98 too close to 1). Quantify Cu with the L-line instead.
- The **Au** peaks come from the substrate under a thin film: a classic quantification trap.

EDX in practice: point, line, map, and the error bars



Point / area

Composition at a spot or averaged over a frame.

Line scan

Composition along a line, e.g. across a film cross-section.

Map

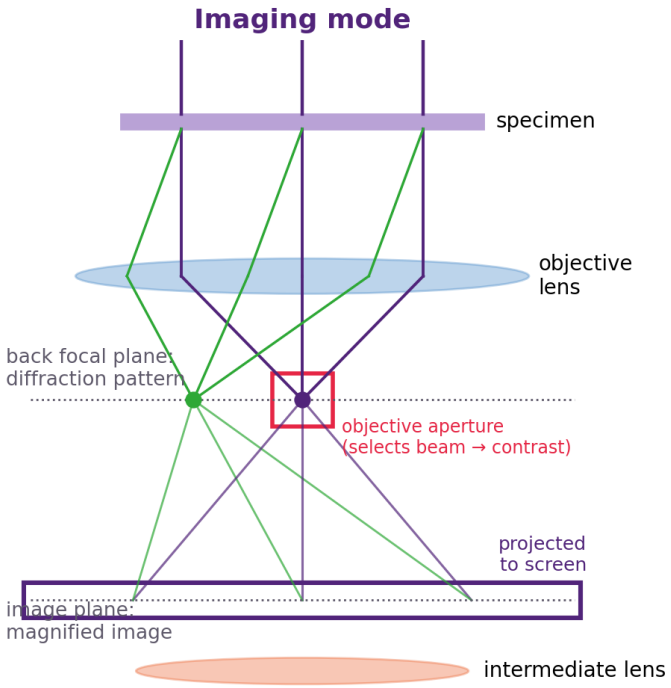
Where each element sits: uniform or segregated?

Limits to quote honestly

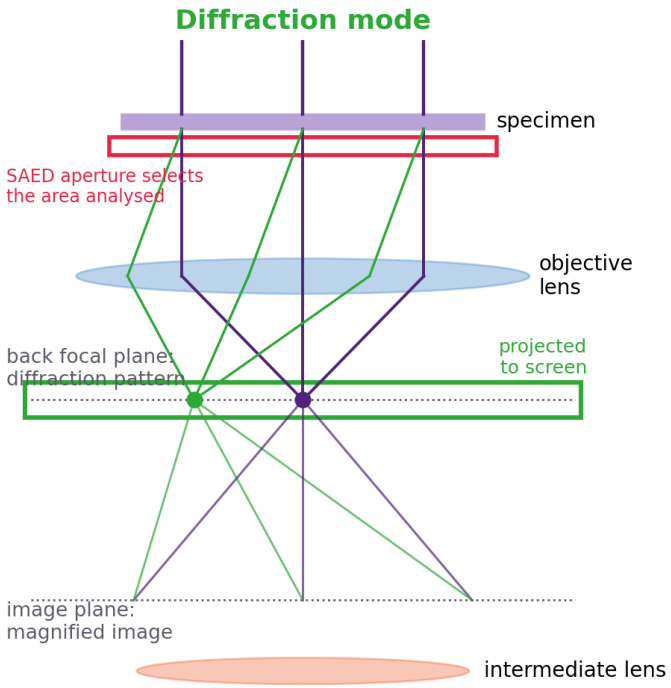
- Spatial resolution ≈ interaction volume: ~1 µm in bulk SEM, ~nm in STEM on thin lamellae
- Typical accuracy ±1–2 at% (standardless); detection limit ~0.1–1 wt%
- Light elements (C, N, O) unreliable
- Peak overlaps; substrate signal from thin films

Cross-check with XPS, which probes only the top few nm. EDX samples the bulk of the film; XPS sees the surface chemistry (oxides, contamination). The case study reports both.

TEM: electrons through a thin slice



BF image / HRTEM lattice fringes



SAED pattern (spots / rings)

$\lambda = h/\sqrt{[2m_0eV(1 + eV/2m_0c^2)]} \rightarrow 2.51$ pm (200 kV)

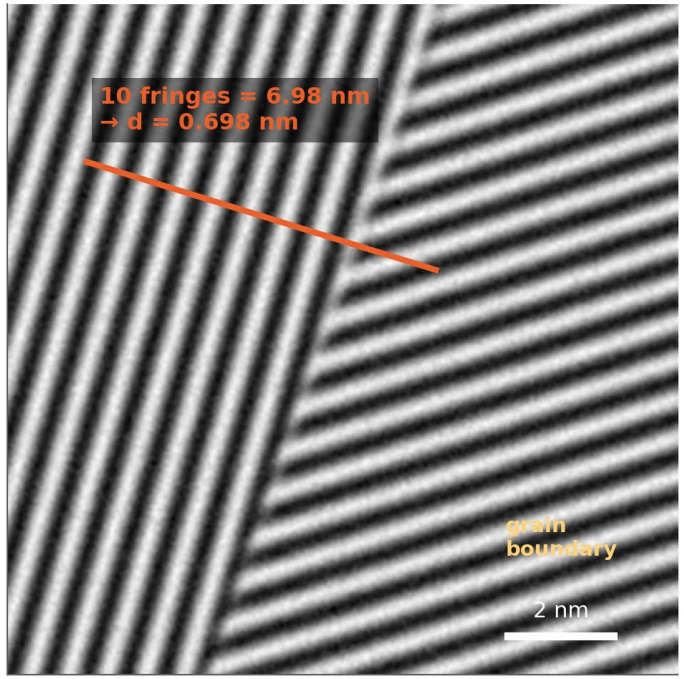
- **Specimen < 100 nm thick**: drop-cast nanoparticles, or a FIB-cut lamella for films
- **Imaging mode**: bright field (mass–thickness and diffraction contrast) and HRTEM lattice fringes
- **Diffraction mode**: SAED from a selected area
- **STEM**: scan a fine probe; HAADF intensity $\propto Z^{\sim 1.7}$, plus STEM-EDX maps at nm resolution

One instrument, two pictures of the same region: the real-space image and its diffraction pattern. Switching modes changes which plane the intermediate lens projects.

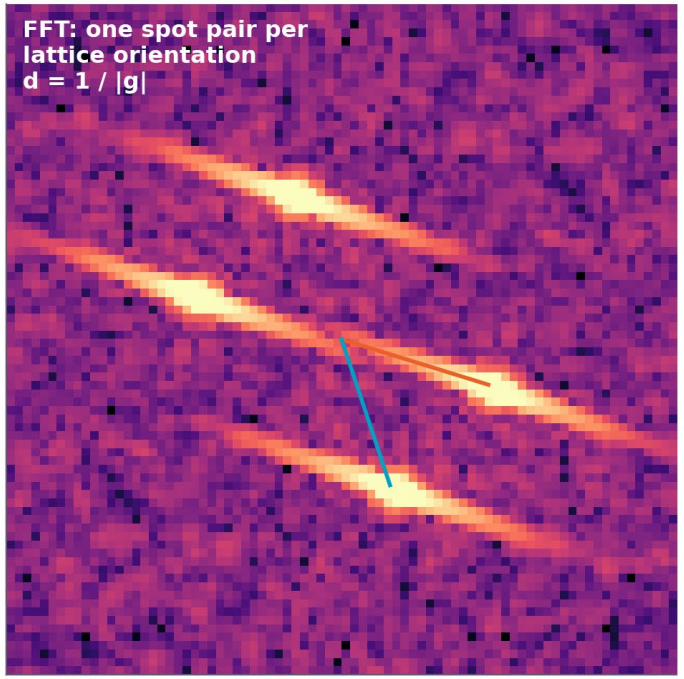
Reading TEM data: lattice fringes and their FFT

B · HRTEM

HRTEM: lattice fringes in two domains



FFT (digital diffractogram)



Measure d from fringes

Measure across 10 fringes and divide by 10: here 6.98 nm / 10 = 0.698 nm.

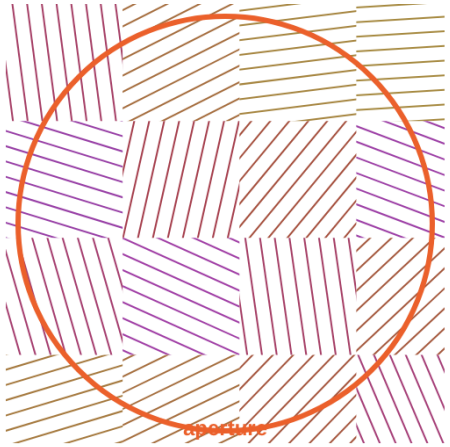
FFT = digital diffraction

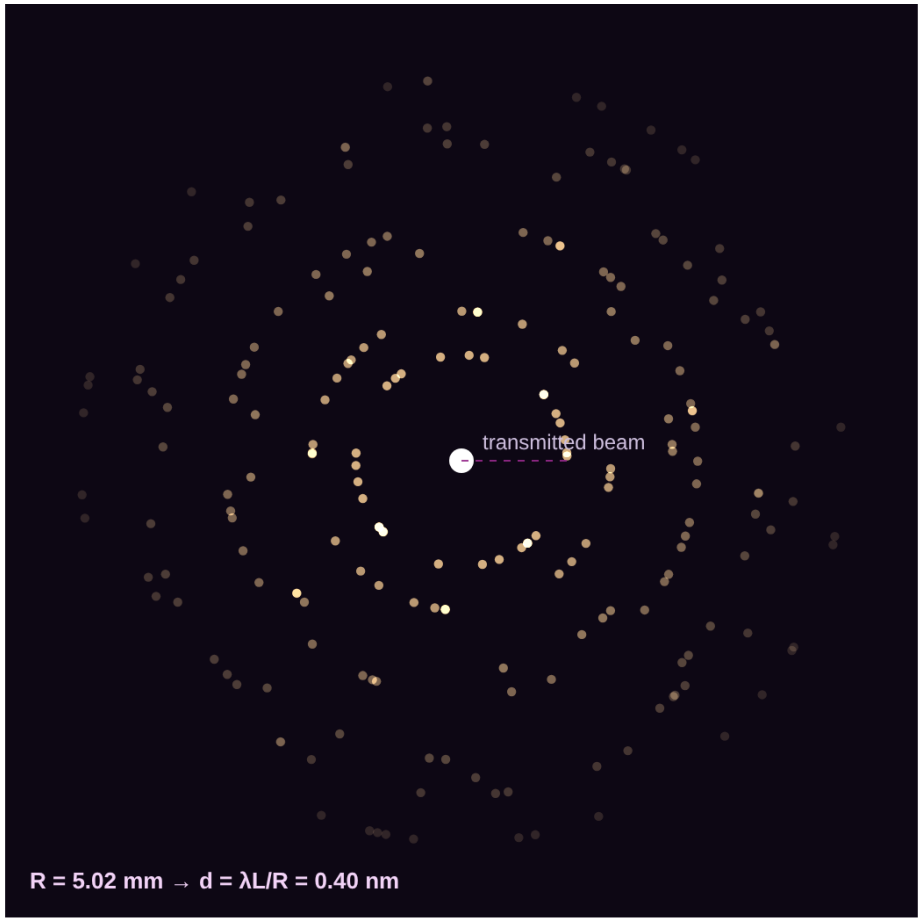
One spot pair per lattice orientation, at |g| = 1/d. Two domains → two spot pairs.

SAED camera constant

R·d = λL: measure the spot or ring radius R on the pattern to get d.

Simulation: spots or rings? Grains inside the aperture



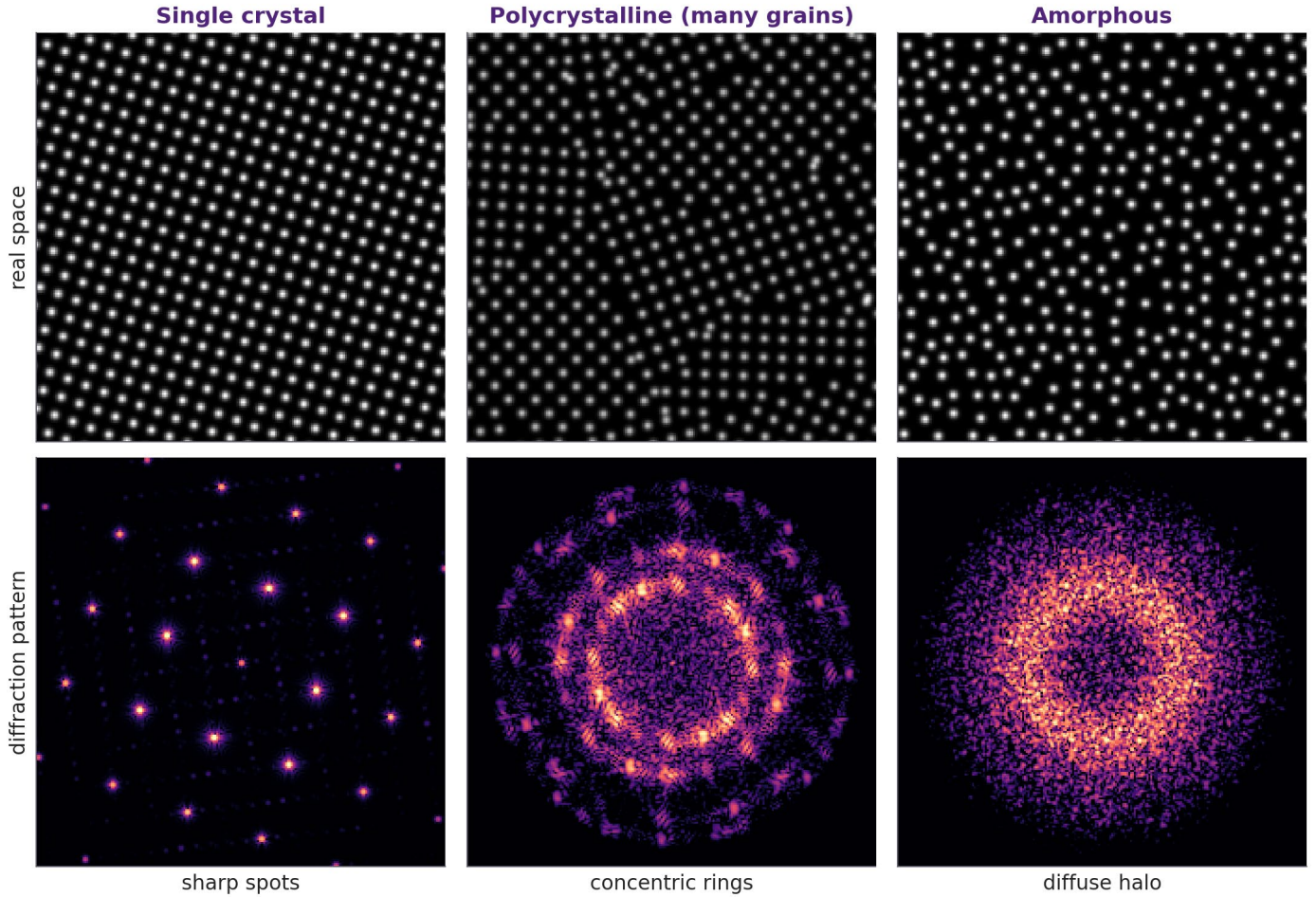


- Small aperture over large grains: **one spot pattern**.
- Widen the aperture or shrink the grains: spots fill in to **rings**.
- Tick amorphous: diffuse **halos**.
- Readout: d = λL/R for the first ring.

Diffraction patterns are a crystallinity fingerprint

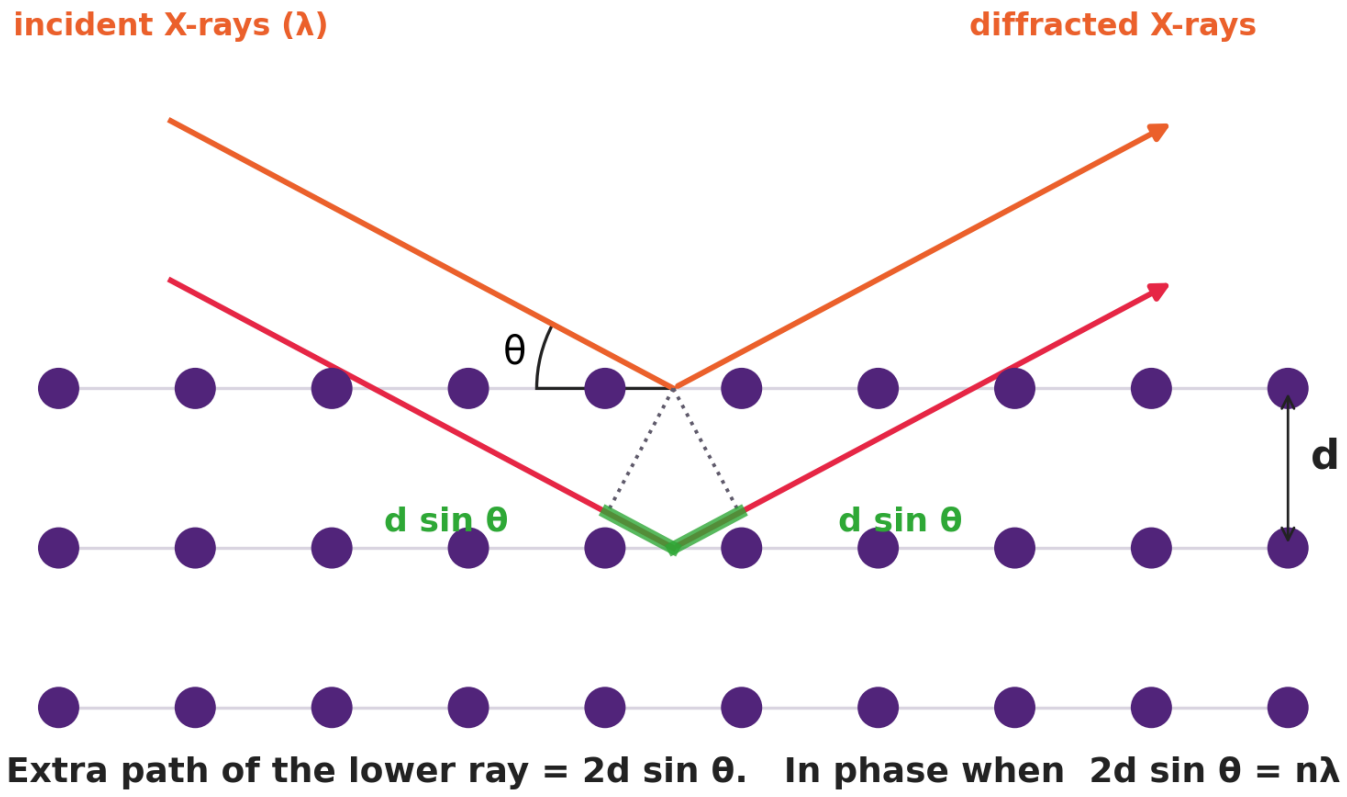
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B · COMPUTED PATTERNS (FFT OF MODEL STRUCTURES)



Single crystal → spots · polycrystal → rings · amorphous → diffuse halo. The same three outcomes appear in SAED and in XRD: sharp, ring-like, or a broad hump.

XRD: Bragg’s law



$$2d \sin \theta = n\lambda$$

Source

Cu Kα, λ = 0.15406 nm (Mo Kα 0.07093 nm for heavy absorbers).

Scan

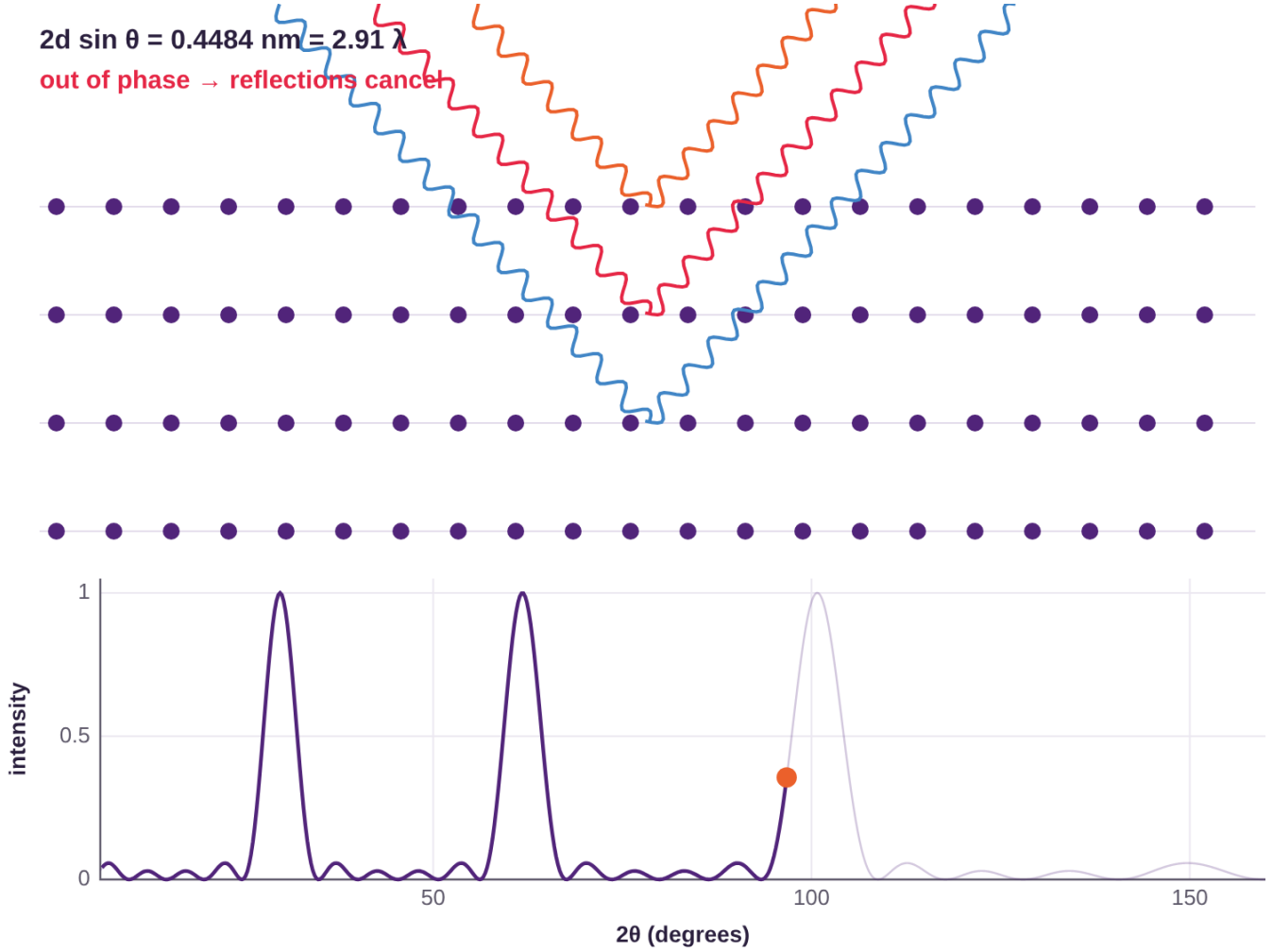
Rotate sample and detector, record intensity versus 2θ. Each peak gives one d-spacing.

Identify

Match the set of d values and intensities to a reference pattern (ICDD PDF, COD) to identify the phase.

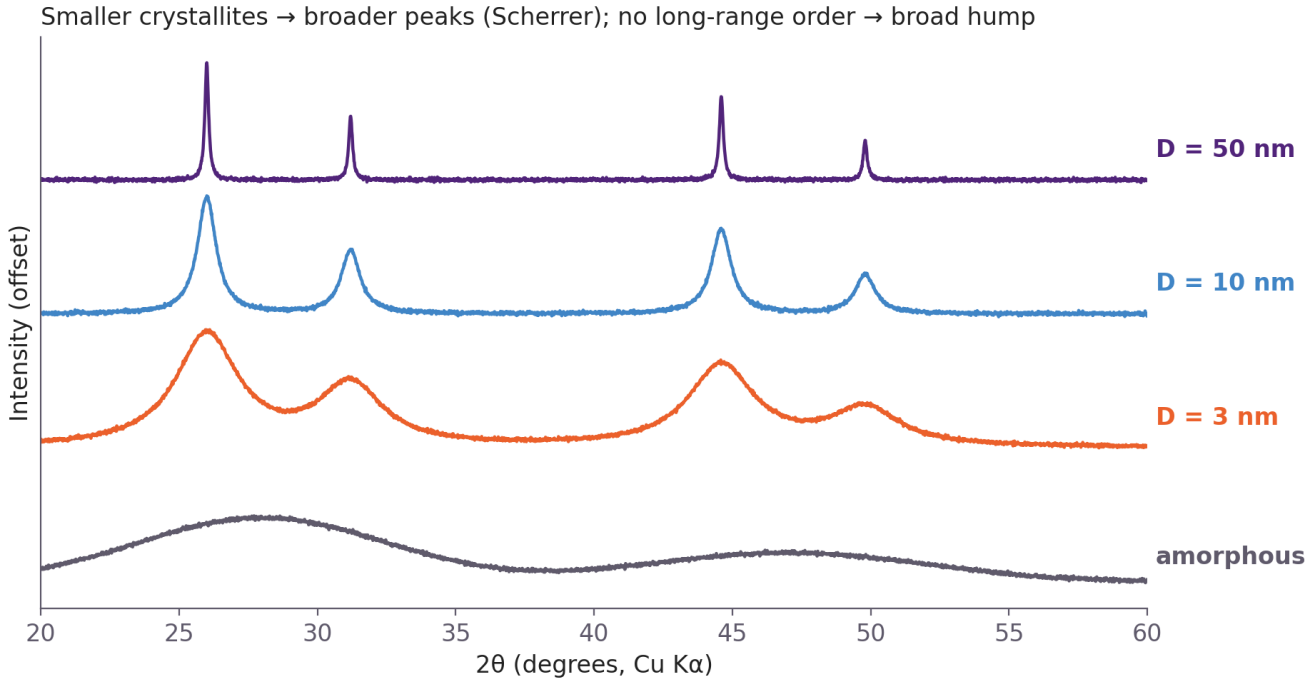
Simulation: sweep θ and watch the Bragg peak appear

B · SIMULATION 14



- **Sweep θ**: waves from successive planes are drawn with their phases.
- When 2d sin θ = nλ the readout turns green.
- Increase the number of planes **N**: the peak sharpens.
- That is the origin of **Scherrer broadening**: few planes, less selective interference, broad peaks.

Reading an XRD pattern: position, intensity, width



Position → phase, d-spacing

Shifts indicate strain or alloying.

Intensity → texture

Ratios that differ from the reference mean preferred orientation.

Width → size and strain

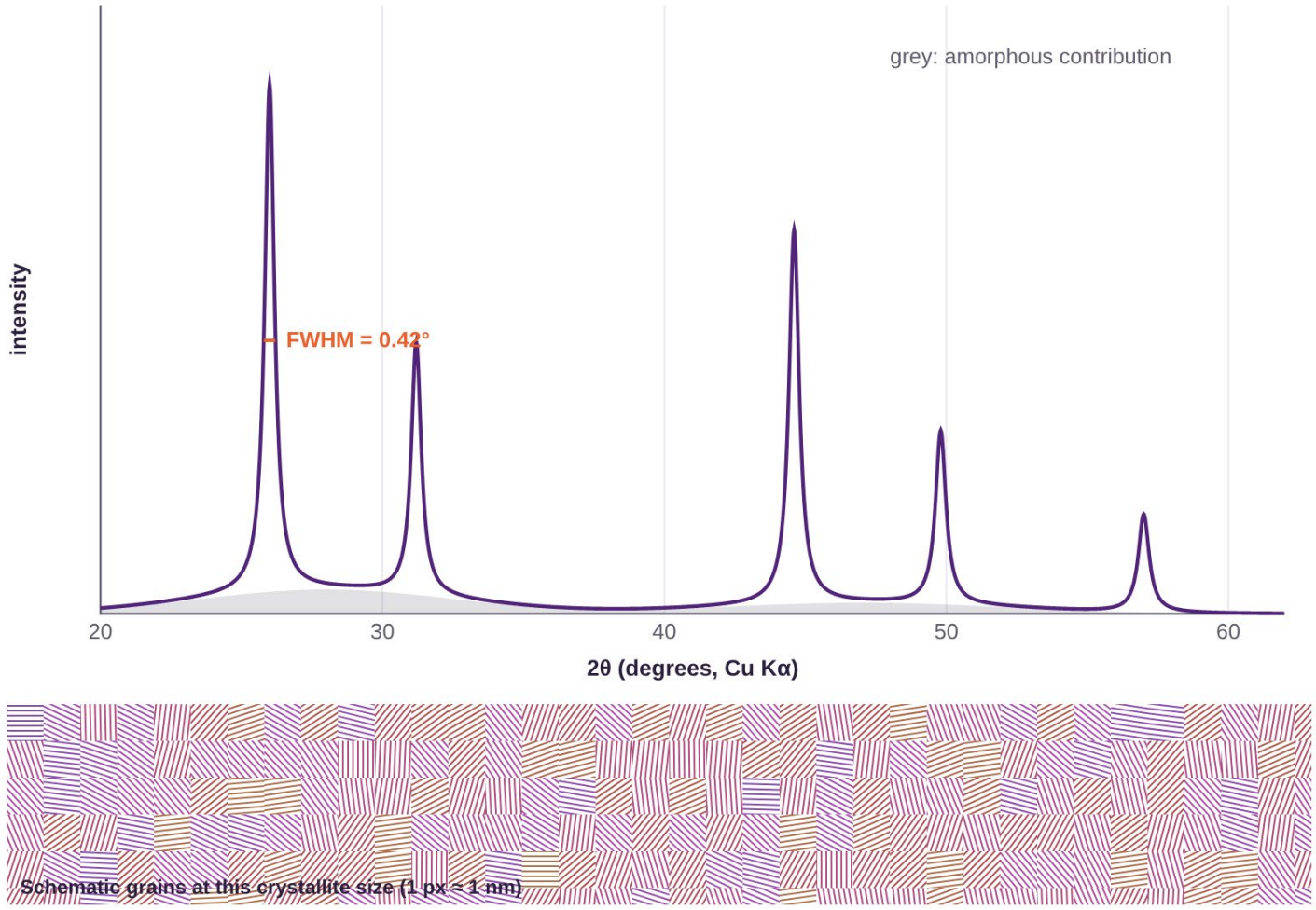
$$D = K\lambda / (\beta \cos \theta)$$

K ≈ 0.9; β = FWHM in radians, instrument-corrected. Valid ≈ 2–200 nm. Crystallite (domain) size, not particle size.

Worked: 2θ = 25.0°, β = 0.40° → D = 0.9 × 0.15406 / (0.00698 × cos 12.5°) ≈ **20 nm**.

Simulation: crystallite size, strain and crystallinity in one pattern

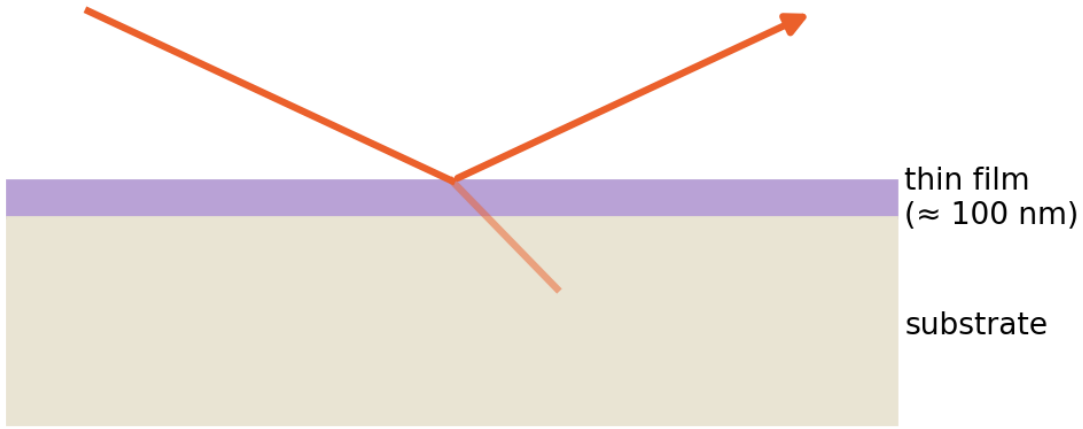
B · SIMULATION 15



- Shrink the crystallite size to **3 nm**: peaks smear into humps (like pristine mesoporous CuTe).
- Add **microstrain**: the Scherrer estimate goes wrong (strain mistaken for small size). Use Williamson–Hall.
- Raise the **amorphous fraction**: the hump grows and crystallinity falls.

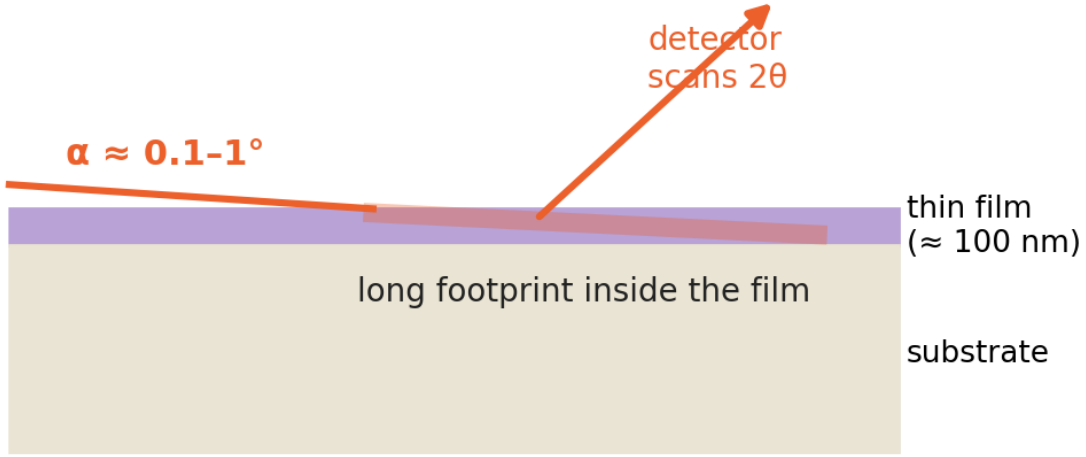
Thin films: grazing-incidence XRD (GIXD)

Symmetric θ–2θ (Bragg–Brentano)



Beam goes straight through the film: substrate peaks dominate

Grazing incidence (GIXD)



Path stays in the film: film peaks enhanced, substrate suppressed

Problem

A 100 nm film diffracts weakly; in θ–2θ the beam penetrates microns and the substrate dominates.

Fix

Keep the incident angle near the critical angle (0.1–1°): a long footprint in the film, little substrate signal.

Out-of-plane vs in-plane

Choose which lattice planes you probe: parallel or perpendicular to the substrate. Useful for oriented films.

Which technique, for which question?

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B · SUMMARY TABLE

	Probe → signal	Tells you	Resolution / sampling	Sample needs	Blind to
SEM	e^- (1–30 kV) → SE, BSE	morphology, pores, thickness	~1–5 nm; local	vacuum-stable, conductive or coated	crystal structure
EDX	e^- → characteristic X-rays	elements, ratios, maps	~1 µm (SEM), ~1 nm (STEM)	flat, enough overvoltage	phase, oxidation state, light elements
TEM	e^- (80–300 kV) → transmitted, diffracted	grains, d-spacing, defects, SAED	~0.1 nm; very local	< 100 nm thick (FIB lamella)	statistics over large areas
XRD	X-rays → diffracted X-rays	phase, crystallite size, strain, texture	Å in d; averages mm²	flat; GIXD for thin films	amorphous detail, local defects
XPS	X-rays → photoelectrons	surface composition, oxidation states	top 3–10 nm	UHV-compatible	bulk below ~10 nm

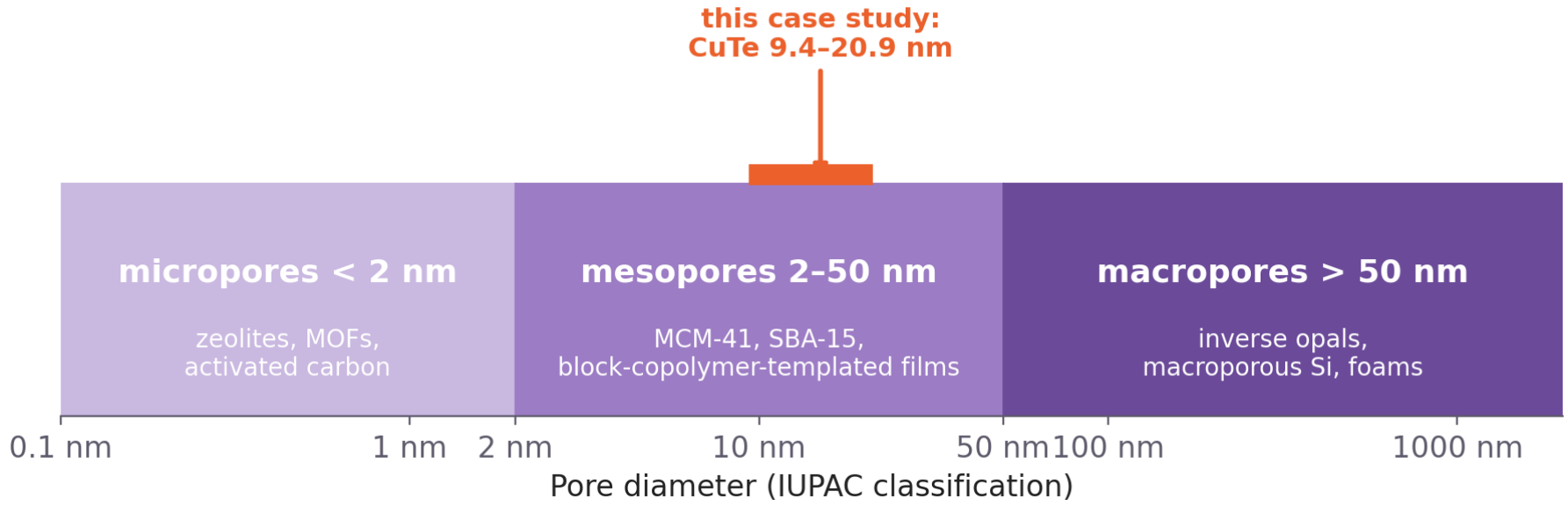
No single technique is enough. A convincing materials claim triangulates: **see it** (SEM/TEM), **identify it** (EDX/XPS), **quantify the order** (XRD/SAED).

C Porous semiconductors

60–68 MINUTES

What a mesoporous semiconductor is, why its pores raise activity, and the price paid in surface states and crystallinity.

What makes a semiconductor porous?



Hard templating

Fill a rigid mould (mesoporous silica, SBA-15/KIT-6) with precursor, then etch the mould away.

Soft templating

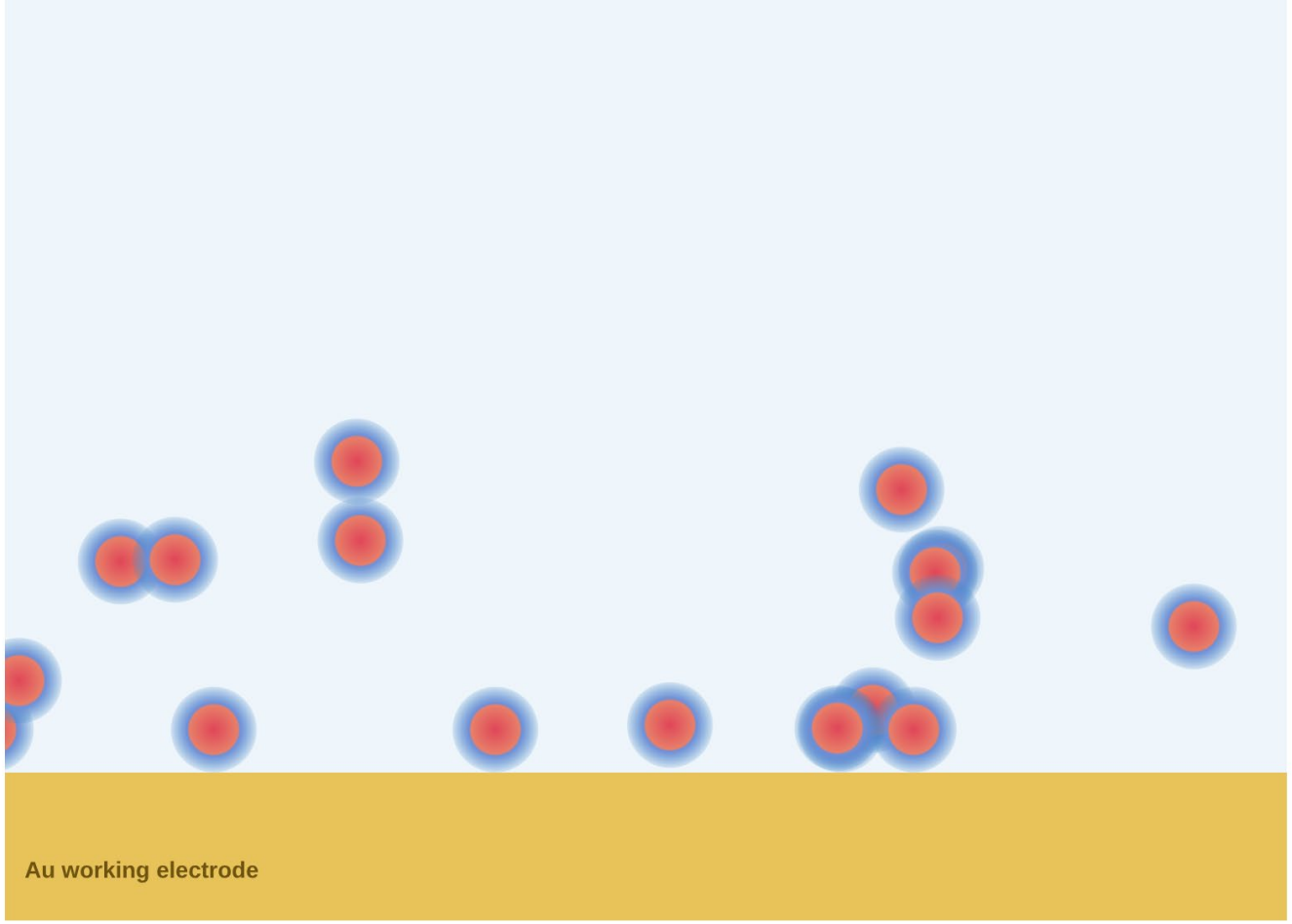
Surfactant or block-copolymer micelles act as removable placeholders during growth (sol–gel, electrodeposition).

Top-down etching

Electrochemical etching (porous Si), dealloying, or anodisation (TiO_2 nanotubes, porous alumina).

Soft templating + electrodeposition, step by step

C · ANIMATION 16



1 Micelles adsorb

PS-*b*-PEO micelles sit on the Au electrode.

2 Electrodeposit

CuTe grows around the micelles.

3 Dissolve

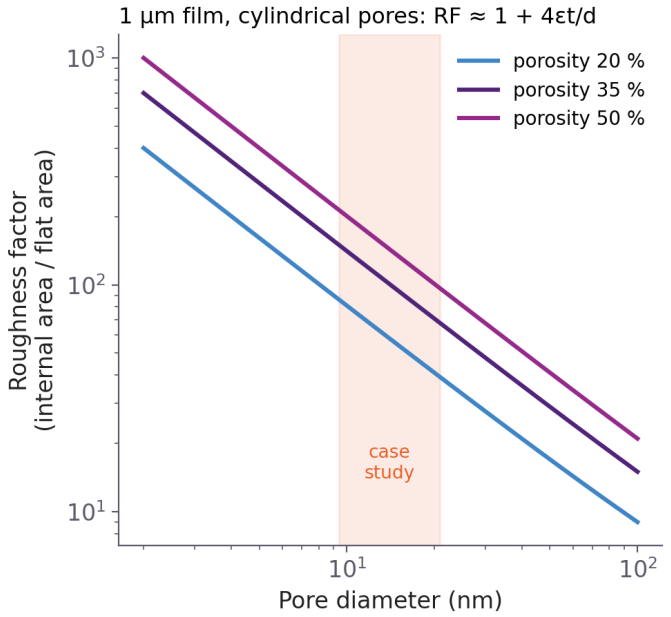
THF removes the polymer.

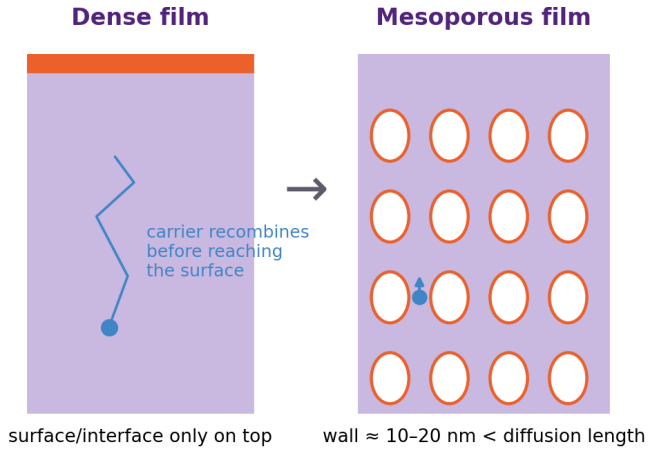
4 Pores remain

Micelle size sets pore size (BCP-1, -2, -3).

Why porosity raises activity

C · FOUR ADVANTAGES





1 · Much more surface

1 µm film, 15 nm pores, 35 % porosity: ~90× the area of a flat film. More active sites.

2 · Short carrier paths

10–20 nm walls are thinner than typical diffusion lengths: carriers reach a surface before recombining.

3 · Light management

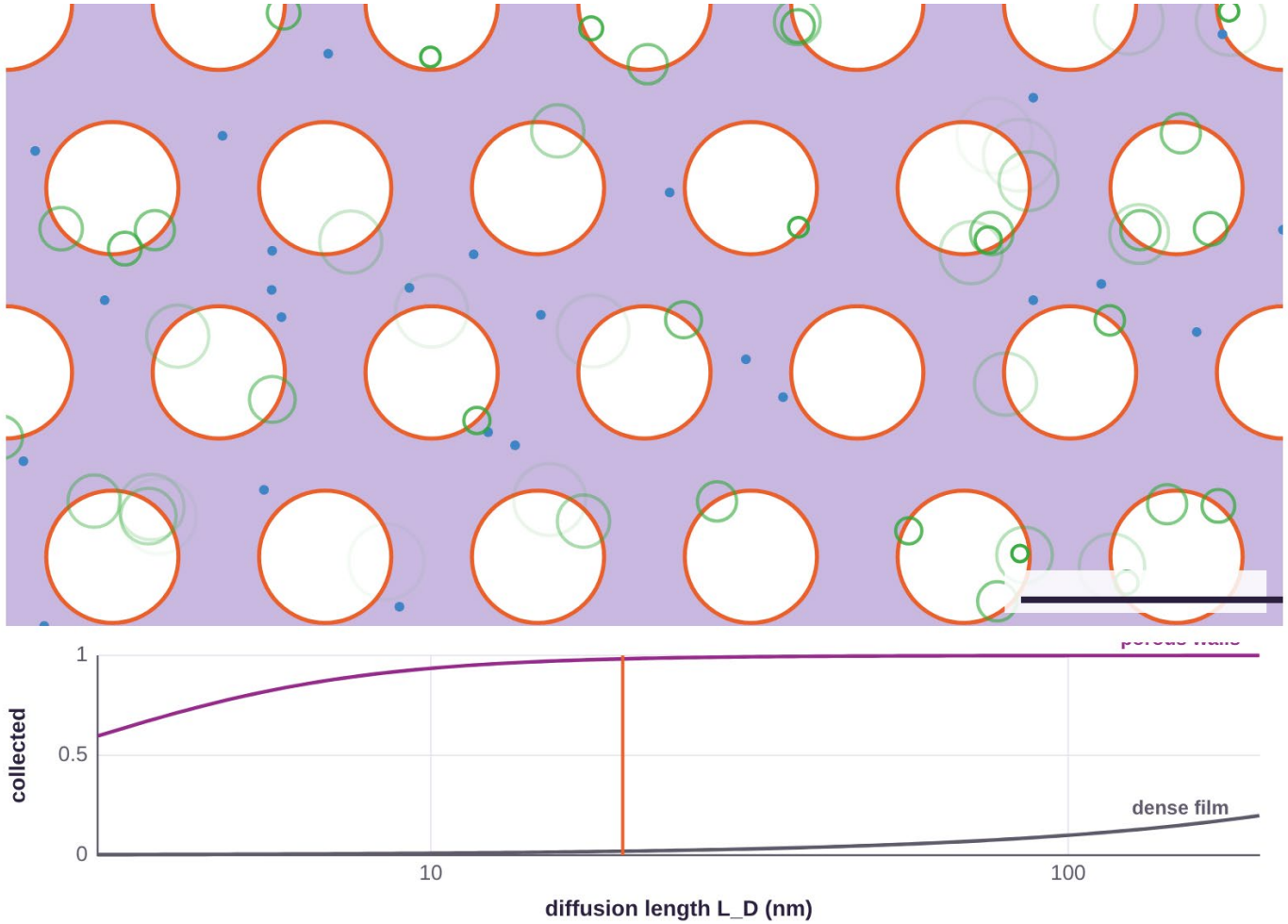
Pores scatter light and lengthen its path; band edges shift with wall size.

4 · Mass transport

Open, interconnected pores let electrolyte and gas molecules in quickly.

Simulation: surface area and carrier collection in a porous film

C · SIMULATION 17



$$RF = 1 + 4\varepsilon t/d$$

- Set L_D = 20 nm: in the porous film almost every carrier reaches a pore surface (green flashes).
- Untick the box: in a dense 1 µm film most carriers recombine before reaching the top surface.
- Slide pore size and porosity: wall thickness and roughness factor change.

The price of porosity: surfaces are also defects

C · WHY DEFECT ENGINEERING IS NEEDED

Dangling bonds and surface states

Unsaturated bonds at pore walls create states in the gap: carrier traps, recombination centres, Fermi-level pinning, slow response.

Small, strained domains

Growth confined around micelles gives nanocrystalline walls with many domain boundaries: lower crystallinity, broad XRD peaks.

Band tailing

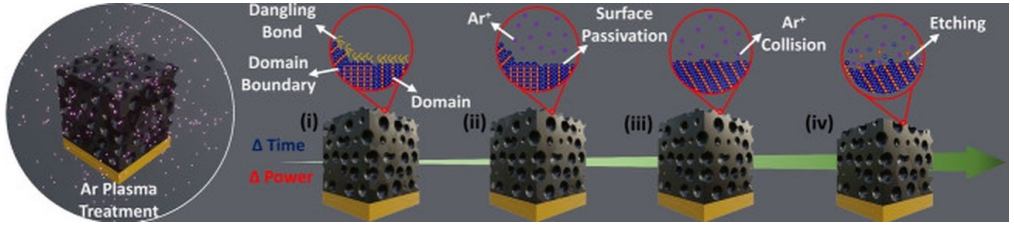
Disorder broadens the absorption edge: a larger Urbach energy E_U ($\alpha = \alpha_0 \cdot \exp(h\nu/E_U)$).

Surface oxidation

More surface means more oxide and contamination.

The usual fix fails. Thermal annealing improves crystallinity, but at a few hundred °C the pores sinter, distort and collapse.

The question for the case study: can we grow the domains and passivate the surfaces **without** destroying the pores?



Graphical abstract: Ashok et al., Angew. Chem. Int. Ed. 2022, 61, e202114729 (CC BY-NC 4.0).

D Case study: plasma domain engineering

68–84 MINUTES

Mesoporous CuTe films, one low-power Ar plasma step, and what SEM, EDX, TEM, XRD and XPS each revealed.

The paper and the question

D · ASHOK ET AL., ANGEW. CHEM. INT. ED. 2022

Plasma-Induced Nanocrystalline Domain Engineering and Surface Passivation in Mesoporous Chalcogenide Semiconductor Thin Films

A. Ashok, A. Vasanth, T. Nagaura, M. Eguchi, N. Motta, H.-P. Phan, N.-T. Nguyen, J. G. Shapter, J. Na, Y. Yamauchi. Angew. Chem. Int. Ed. 2022, 61, e202114729. doi:10.1002/anie.202114729 (open access, CC BY-NC 4.0)

Material

Mesoporous copper telluride (CuTe, orthorhombic vulcanite), electrodeposited on Au with PS-b-PEO micelle templates. Three pore sizes plus a non-porous reference.

Treatment

Low-power Ar plasma (13.56 MHz RF), optimum **30 W for 180 s**, substrate at about 23–35 °C. Benchmarked against annealing at 70, 140 and 210 °C.

Claim

Larger, aligned crystalline domains and passivated surfaces, with the pores intact: crystallinity ~70 % → ~85 %.

85 %

crystallinity after plasma (from ~70 %)

180 s at 30 W

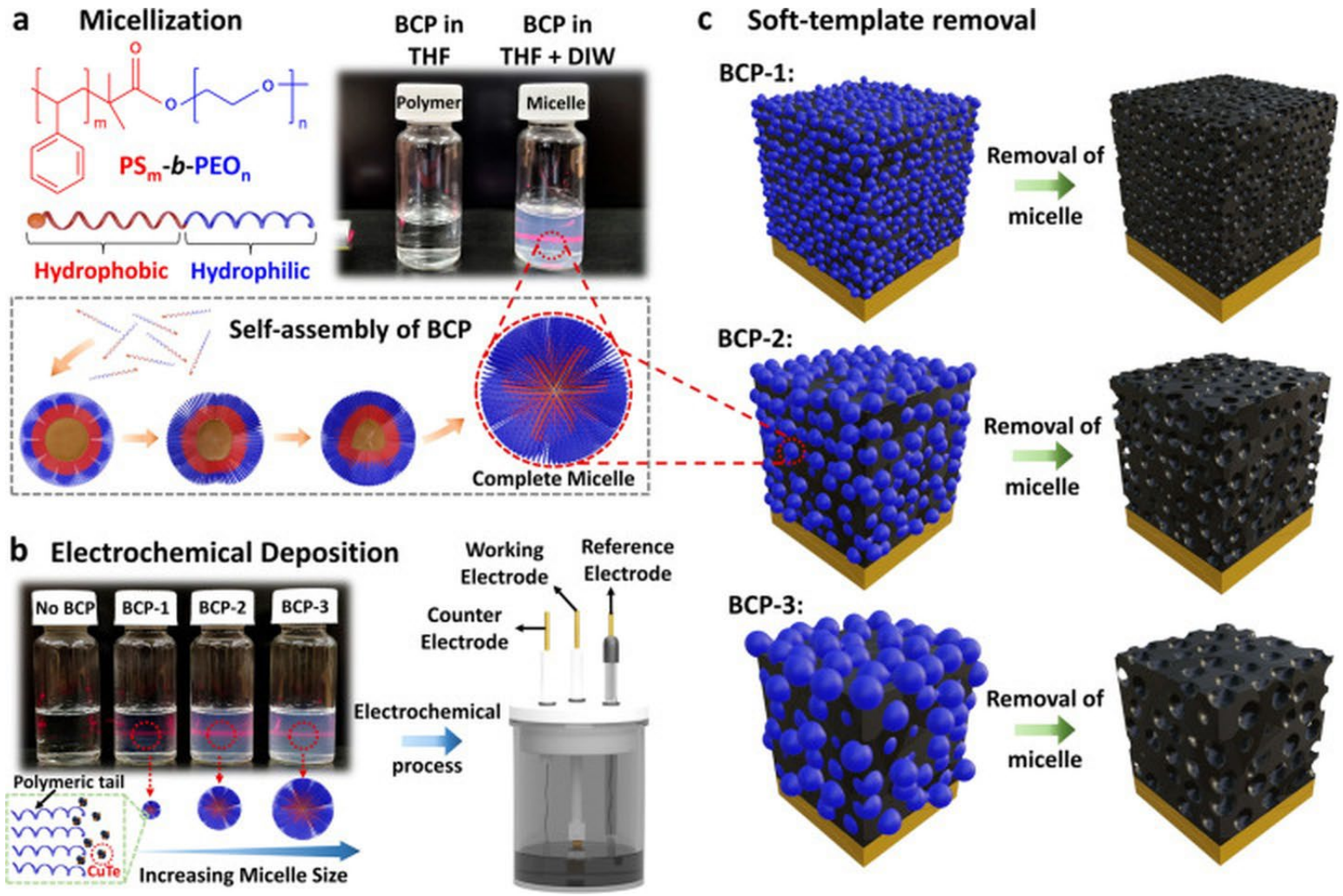
the whole treatment, near room temperature

≥ 140 °C

annealing temperature at which pores distorted and collapsed

Step 1 · Making the films: micelles + electrodeposition

D · FIG. 1



a · Micellisation

PS$_m$-b-PEO$_n$ in THF; adding water forms micelles (hydrophobic PS core, hydrophilic PEO corona).

b · Electrodeposition

Cu and Te precursors with micelles; −0.2 V vs Ag/AgCl; growth ~0.2 nm/s. Polymer molecular weight sets micelle size: BCP-1, -2, -3.

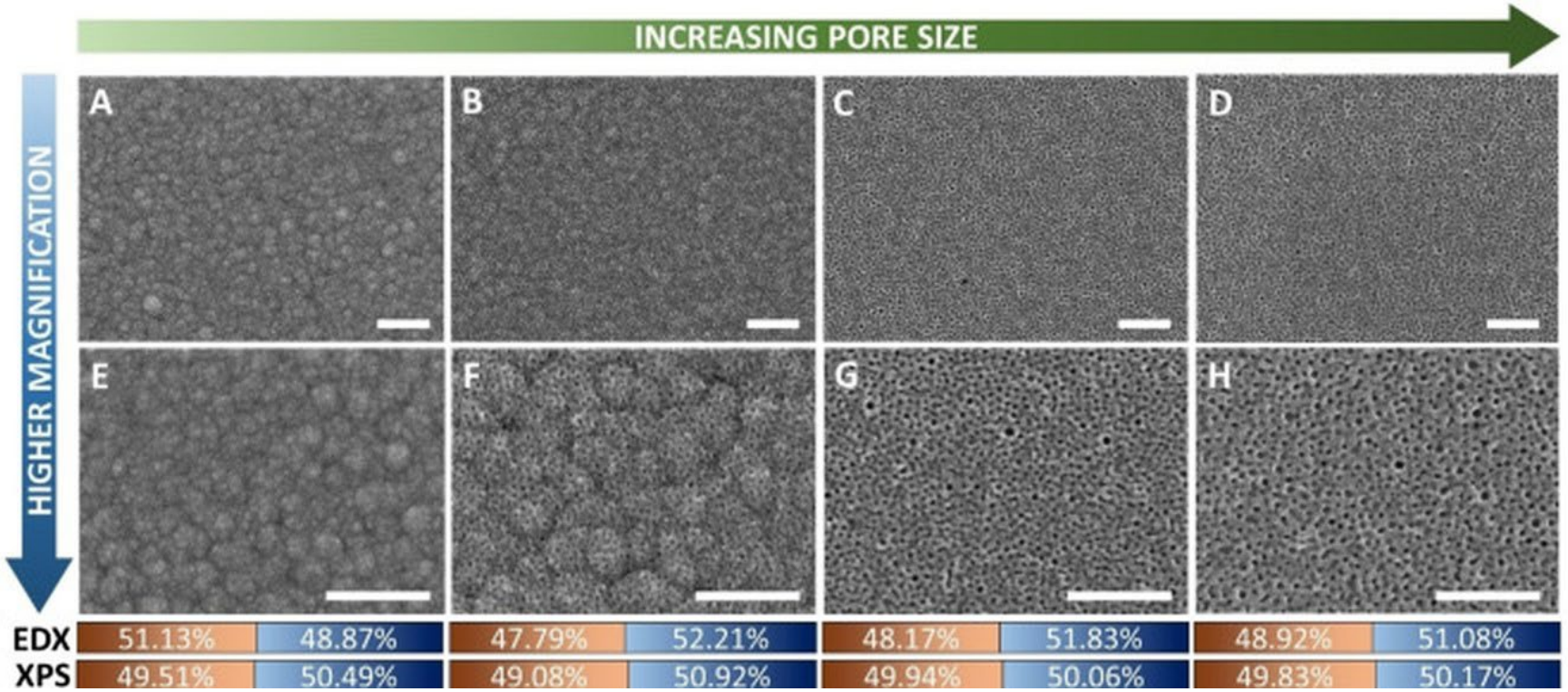
c · Template removal

THF dissolves the micelles, leaving mesopores.

Figure: Ashok et al., Angew. Chem. Int. Ed. 2022, 61, e202114729, Fig. 1 (CC BY-NC 4.0).

Step 2 · SEM and EDX: pores as designed, composition near 1 : 1

D · FIG. 2A–H · SEM · EDX · XPS



9.4 nm
BCP-1 pores · walls 9.2 nm

16.2 nm
BCP-2 pores · walls 14.8 nm

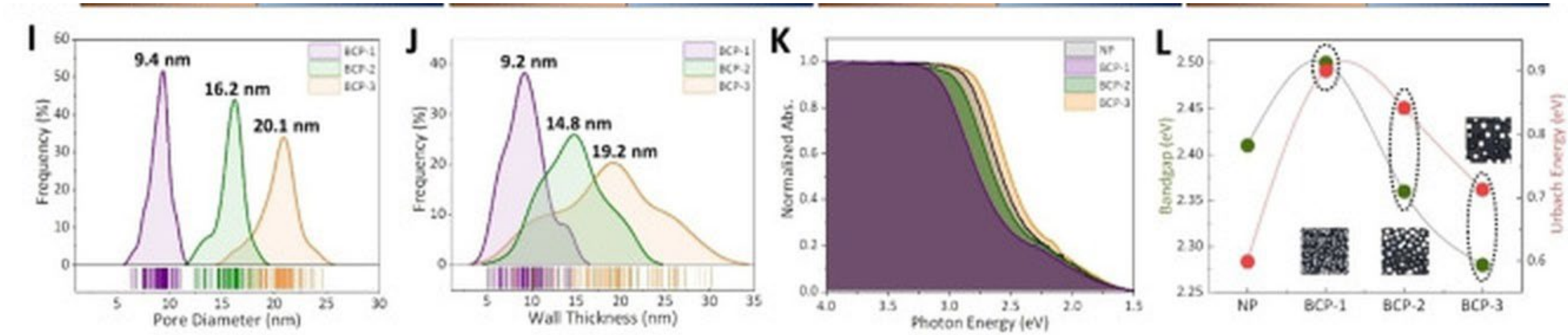
20.9 nm
BCP-3 pores · walls 19.2 nm

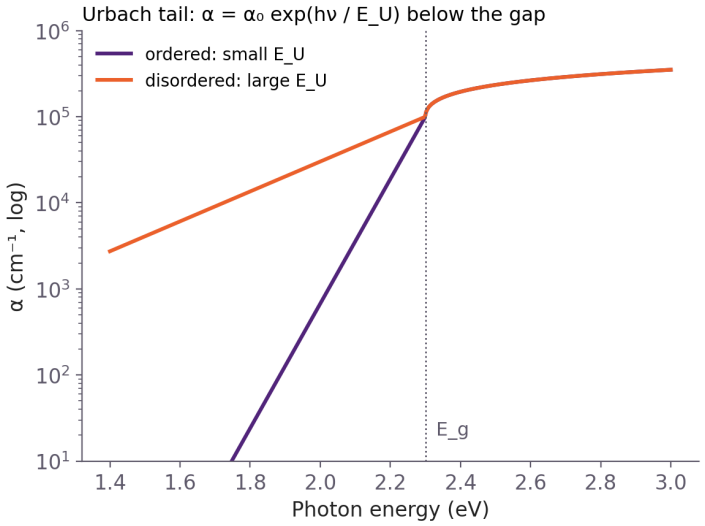
≈ 48 : 52
Cu : Te by EDX (at%); XPS ≈ 49 : 51

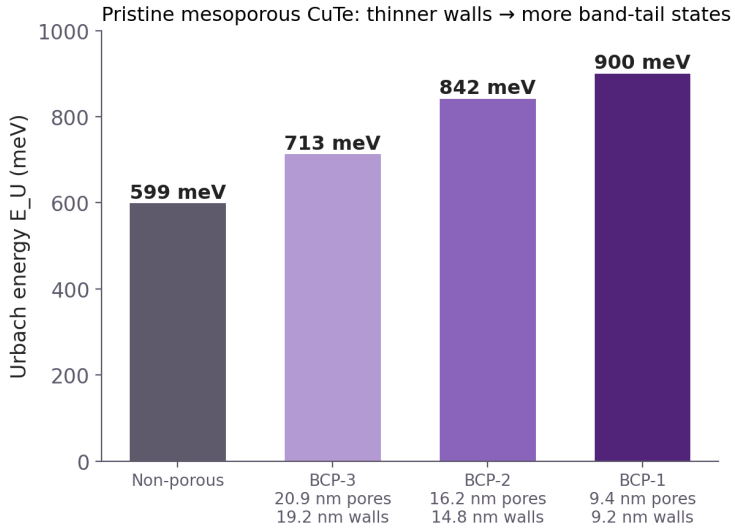
Scale bars 250 nm. Figure: Ashok et al. 2022, Fig. 2 (CC BY-NC 4.0). The text reports 20.9 nm for BCP-3; Fig. 2I labels 20.1 nm.

Step 3 · Thinner walls, more disorder: the Urbach tail

D · FIG. 2I–L · UV–VIS







Reading it

- Absorbance blue-shifts as pores and walls shrink.
- E_U: 599 meV (non-porous) → 713, 842, 900 meV as walls thin from 19 to 9 nm.
- Larger E_U = more tail states (surfaces, strain, small domains).
- After plasma: E_U fell; E_g changed little.

Top: Ashok et al. 2022, Fig. 2I–L (CC BY-NC 4.0). Bottom: Urbach energies replotted from the paper.

Step 4 · Low-power Ar plasma: passivate, crystallise, but do not etch

D · FIG. 3

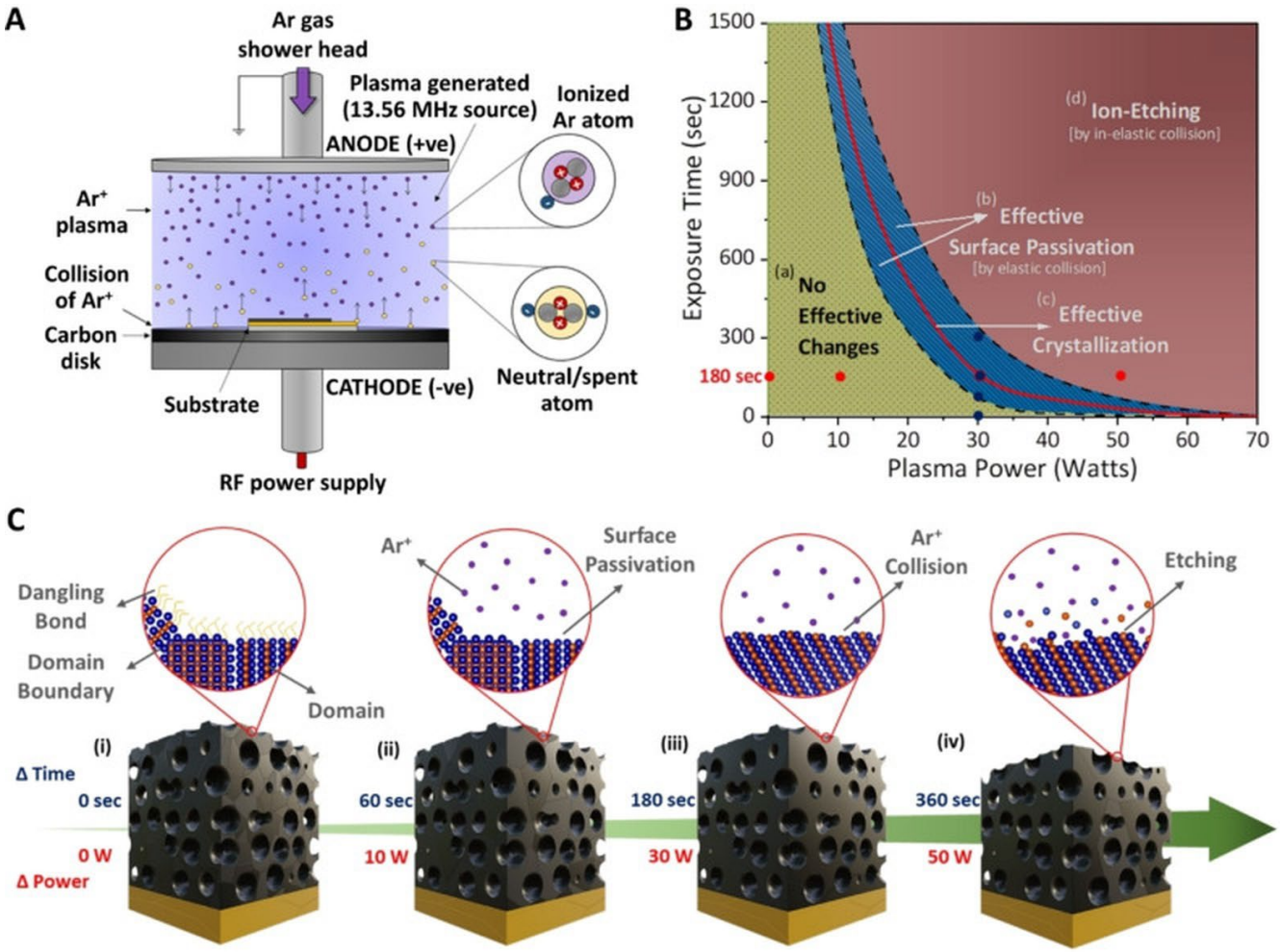


Figure: Ashok et al., Angew. Chem. Int. Ed. 2022, Fig. 3 (CC BY-NC 4.0).

A · Chamber

Parallel-plate RF (13.56 MHz) Ar plasma; Ar^+ accelerated through the sheath to the sample on the cathode.

B · Process map

(a) no change at low power · (b) surface passivation (elastic collisions) · (c) crystallisation · (d) ion etching (inelastic). Optimum **30 W, 180 s**.

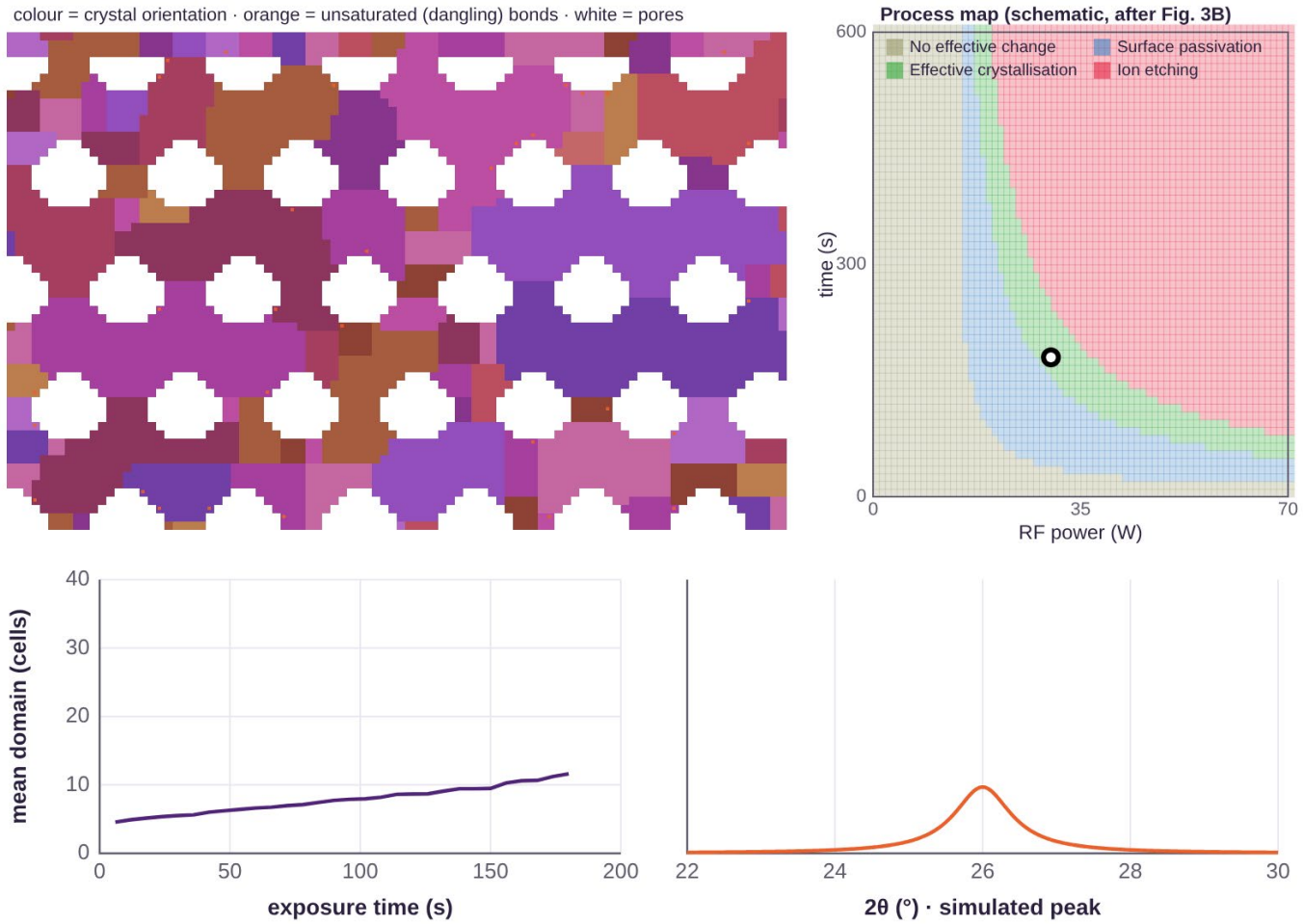
C · What happens

Dangling bonds passivated → domains rearranged and enlarged → over-exposure etches the walls.

At 10 W: no change even for long exposures. At 30 W beyond ~240 s: etching, and E_U rises again.

Simulation: explore the plasma process map

D · ILLUSTRATIVE LATTICE MODEL



10 W, 180 s

nothing happens

30 W, 180 s

bonds passivated, domains coarsen, XRD peak sharpens

50 W, 180 s

domains grow but the surface starts to etch

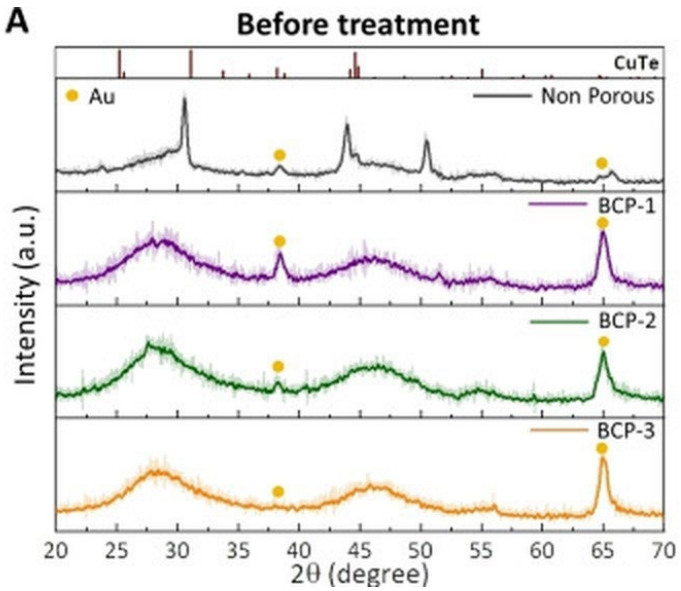
30 W, 360 s

over-exposure etches the walls

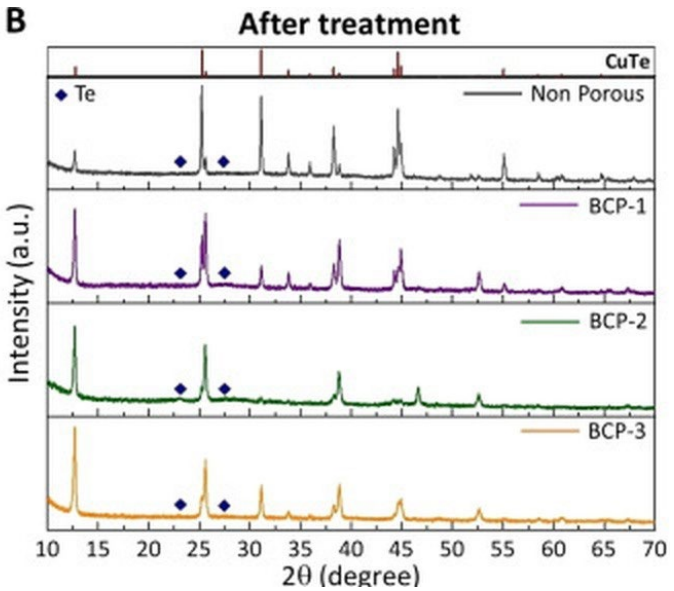
A teaching model tuned to reproduce the regimes of Fig. 3B, not a physical simulation of the plasma.

Step 5 · GIXD: broad humps become sharp peaks

D · FIG. 4A, B · XRD



Before: weak, broad humps in all porous films; sharp Au substrate peaks.



After 30 W, 180 s: many sharp, intense CuTe peaks; small Te peaks (◆) appear.

Width

Broad → sharp: larger coherent domains (Scherrer) and relieved strain. Pristine peaks were too broad to index reliably.

Position

d-spacings changed only slightly and were similar for all pore sizes: same phase, better order.

New peaks

Te reflections: excess Te knocked out of the walls (consistent with the EDX Te excess).

Limit

No further sharpening after 180 s at 30 W; 10 W gave no change.

Figures: Ashok et al. 2022, Fig. 4A,B (CC BY-NC 4.0). Note the different 2θ ranges of the two panels.

Step 6 · XPS: the surface oxides are stripped away

D · FIG. 4C, D · XPS

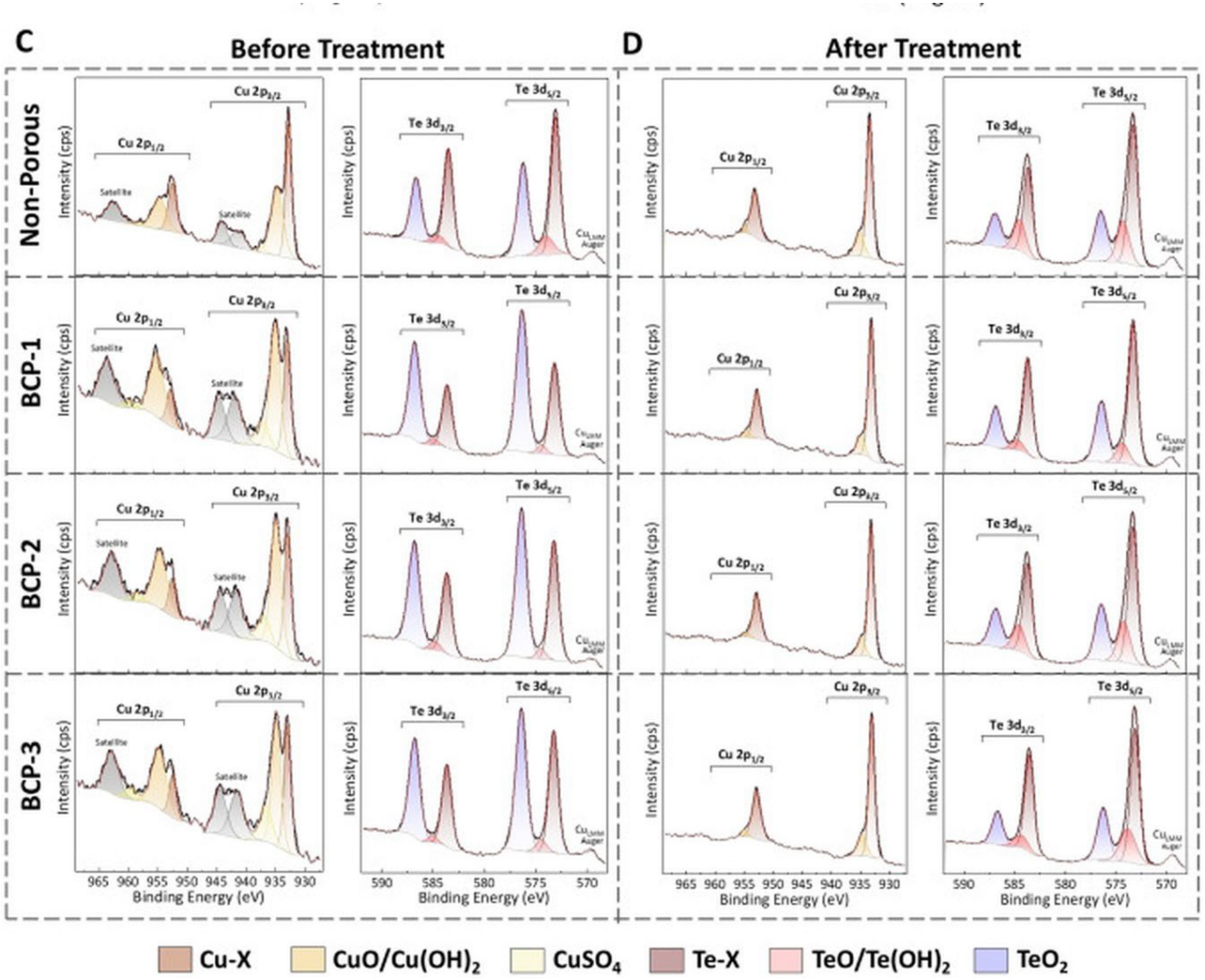


Figure: Ashok et al. 2022, Fig. 4C,D (CC BY-NC 4.0).

Before

Strong Cu 2p satellites and $CuO/Cu(OH)_2$, $CuSO_4$ components; large TeO_2 in Te 3d. Oxide signals are larger in the porous films (more surface).

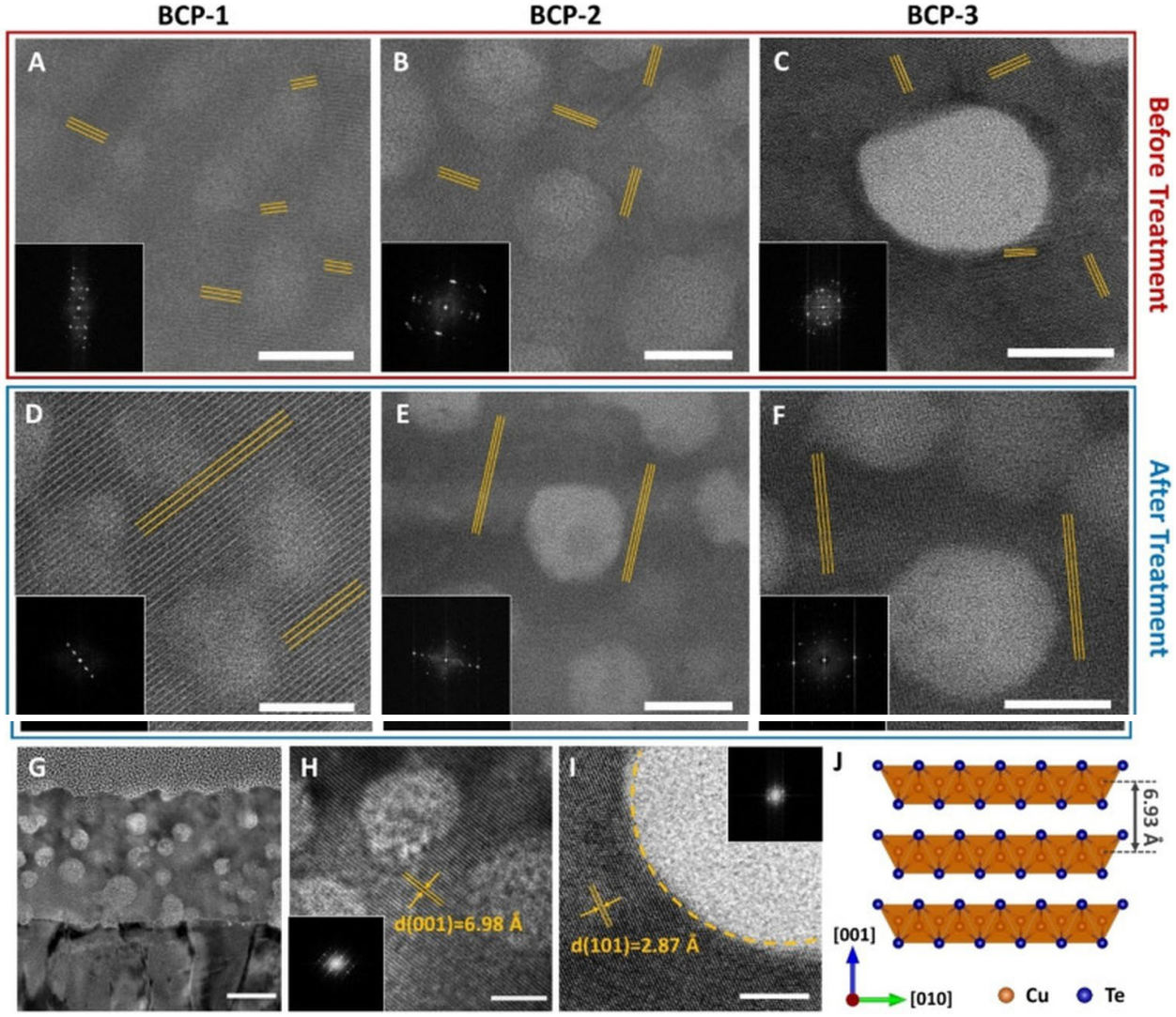
After

Oxide components and satellites largely gone; Cu–Te and Te–Cu dominate.

XPS sees only the top few nm, which is exactly where pores add area. EDX could not show this: it averages through the film and says nothing about oxidation state.

Step 7 · TEM: domains grow and line up

D · FIG. 5 · TEM



Before (STEM-BF, A–C)

Small domains (~5–20 nm) with several orientations (yellow fringe markers); FFT insets show many spots.

After plasma (D–F)

Larger domains with fringes running one way, oriented vertically from the substrate; cleaner FFT.

Lattice (G–J)

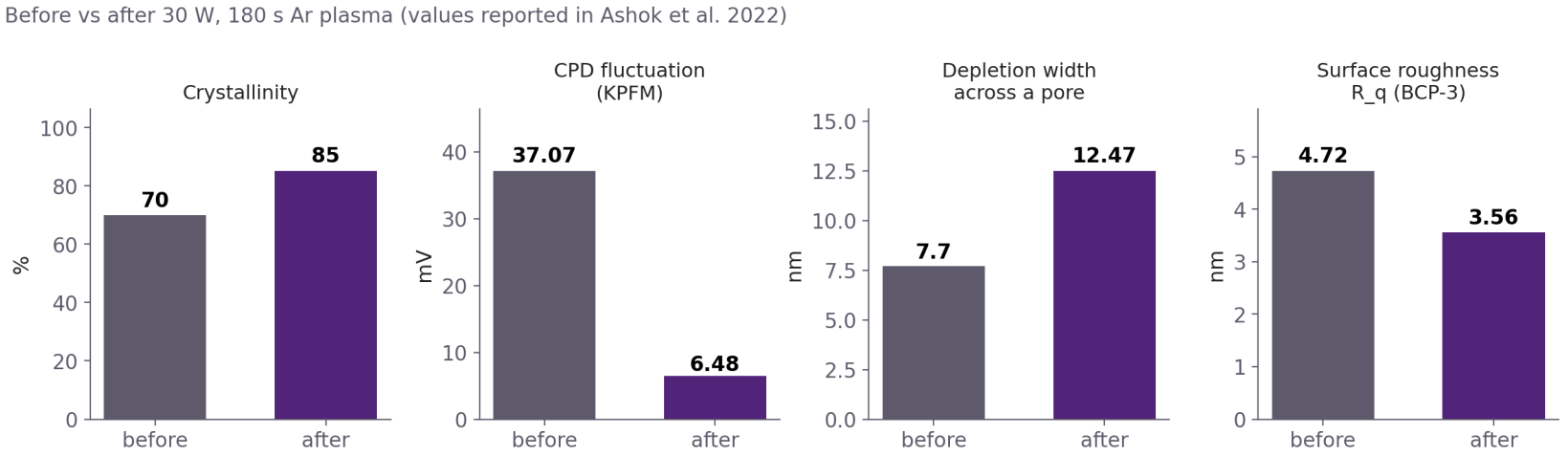
HRTEM: d(001) = 6.98 Å, d(101) = 2.87 Å, vulcanite CuTe.
HAADF: ~2 nm Te clusters on pore surfaces in BCP-3.

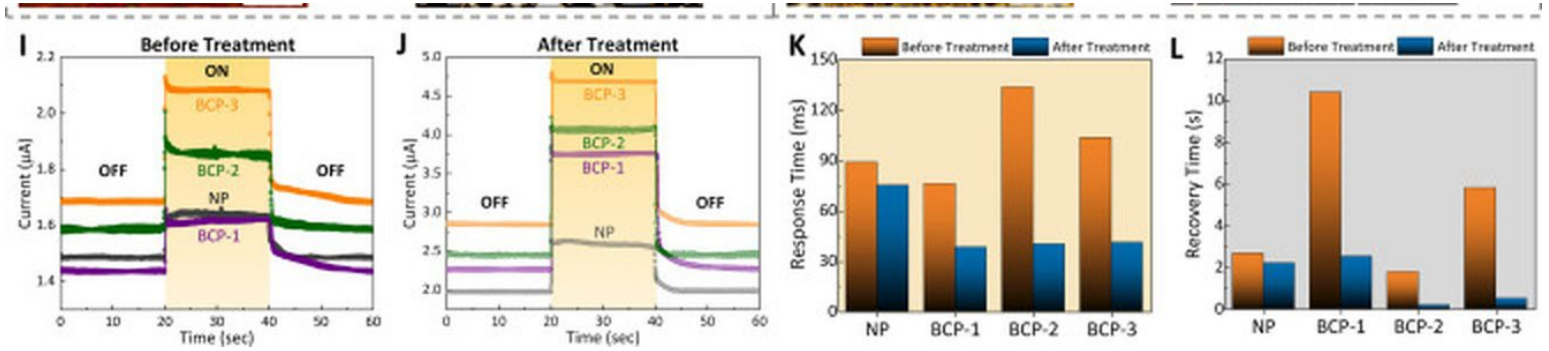
TEM is very local: it must agree with the GIXD average before we believe the claim. Before/after wipe slider: HTML slide 54.

Figures: Ashok et al. 2022, Fig. 5 (CC BY-NC 4.0). Scale bars: A–F 15 nm; G 50 nm; H 10 nm; I 5 nm.

Step 8 · Does it matter? Surface potential and photoresponse

D · FIG. 6 · KPFM / C-AFM + PHOTOELECTROCHEMICAL RESPONSE





~60 % faster response
mesoporous films (non-porous: ~15 %)

~86 % faster recovery
mesoporous films (non-porous: ~18 %)

+30 % light current
porous vs non-porous after plasma (+15 % dark)

Figure: Ashok et al. 2022, Fig. 6 (CC BY-NC 4.0); metrics from the paper text.

What each technique told us

Technique	Before plasma	After 30 W, 180 s	Conclusion
SEM	9.4 / 16.2 / 20.9 nm pores, uniform	pores unchanged (annealing ≥ 140 °C collapsed them)	architecture preserved
EDX	Cu : Te ≈ 1 : 1, slight Te excess	excess Te expelled	stoichiometry kept
UV–vis	E_U 599 → 900 meV as walls thin	E_U reduced; E_g slightly lower	fewer tail states
GIXD	weak, broad humps	sharp, intense peaks; crystallinity ~70 → 85 %	larger domains
XPS	$CuO/Cu(OH)_2$, $CuSO_4$, TeO_2	oxides largely removed	surface passivated
TEM	5–20 nm domains, mixed orientation	larger, aligned domains; d(001) 6.98 Å	local proof of domain growth
KPFM + PEC	CPD fluctuation 37 mV; slow	6.5 mV; response ~60 %, recovery ~86 % faster	better device

Take-home: no single image proves “higher crystallinity, same pores, fewer defects”. SEM + EDX + XRD + TEM + XPS + a device measurement together do.

Summary

E · SIX IDEAS TO KEEP

WHAT IT CAN SEE

$$\lambda_c = 1.24 / E_g$$

Band gap sets the cut-off.

HOW MUCH SIGNAL

$$R = \eta\lambda/1.24 \times G$$

Gain costs bandwidth and adds noise.

HOW FAINT

$$D^* = \sqrt{(A\Delta f)}/NEP$$

With measured noise.

HOW TO SEE STRUCTURE

SEM · EDX · TEM · XRD

Morphology, composition, local order, average order.

WHY PORES

area + short paths

Paid for with surface states.

DEFECT ENGINEERING

30 W · 180 s Ar

Bigger domains, same pores, faster response.

Discussion questions

E · OPEN DISCUSSION · Q&A

1 · Proving “more crystalline”

Which two independent measurements would you insist on before accepting that a treatment increased crystallinity? Why one of each kind (average vs local)?

2 · EDX vs XPS

EDX says Cu : Te ≈ 1 : 1 but XPS shows strong oxides. Can both be right? What does each one sample?

3 · How small a pore?

Smaller pores mean more area but thinner, more defective walls. How would you choose the pore size for a photodetector versus a photocatalyst?

4 · Plasma vs annealing

What other low-temperature routes could crystallise a fragile nanostructure (laser, microwave, chemical passivation)? What would you measure to compare them?

References and resources

E · FURTHER READING

Case study

- A. Ashok et al., Angew. Chem. Int. Ed. 2022, 61, e202114729. doi:10.1002/anie.202114729 (CC BY-NC 4.0; figures reproduced for teaching)

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Characterisation

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Porous materials

- Thommes et al., Pure Appl. Chem. 87, 1051 (2015): IUPAC physisorption and pore classes.
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Questions and discussion

Thank you

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